



IT-OT Communication Networks for Power Grids and Cyber Security

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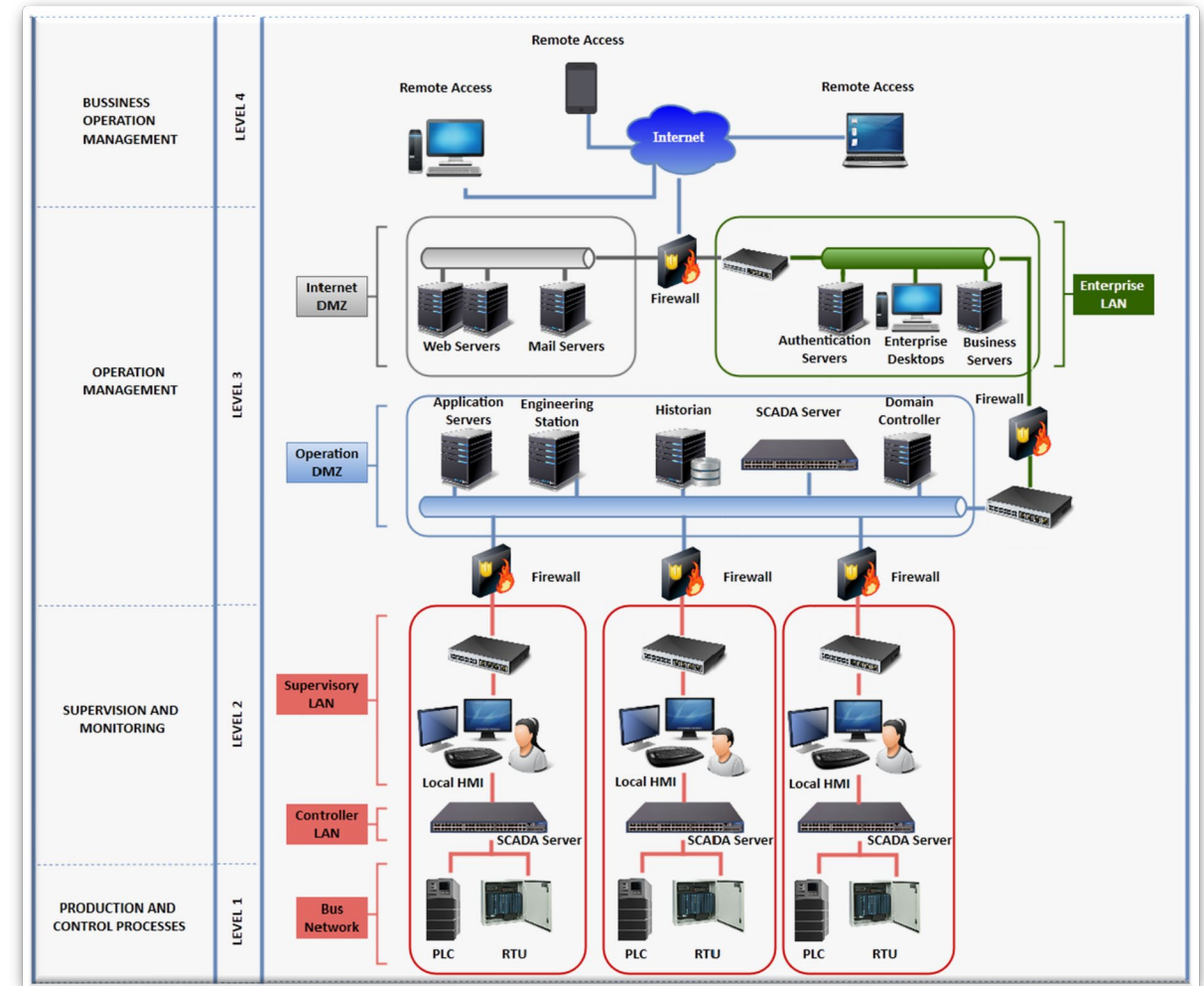
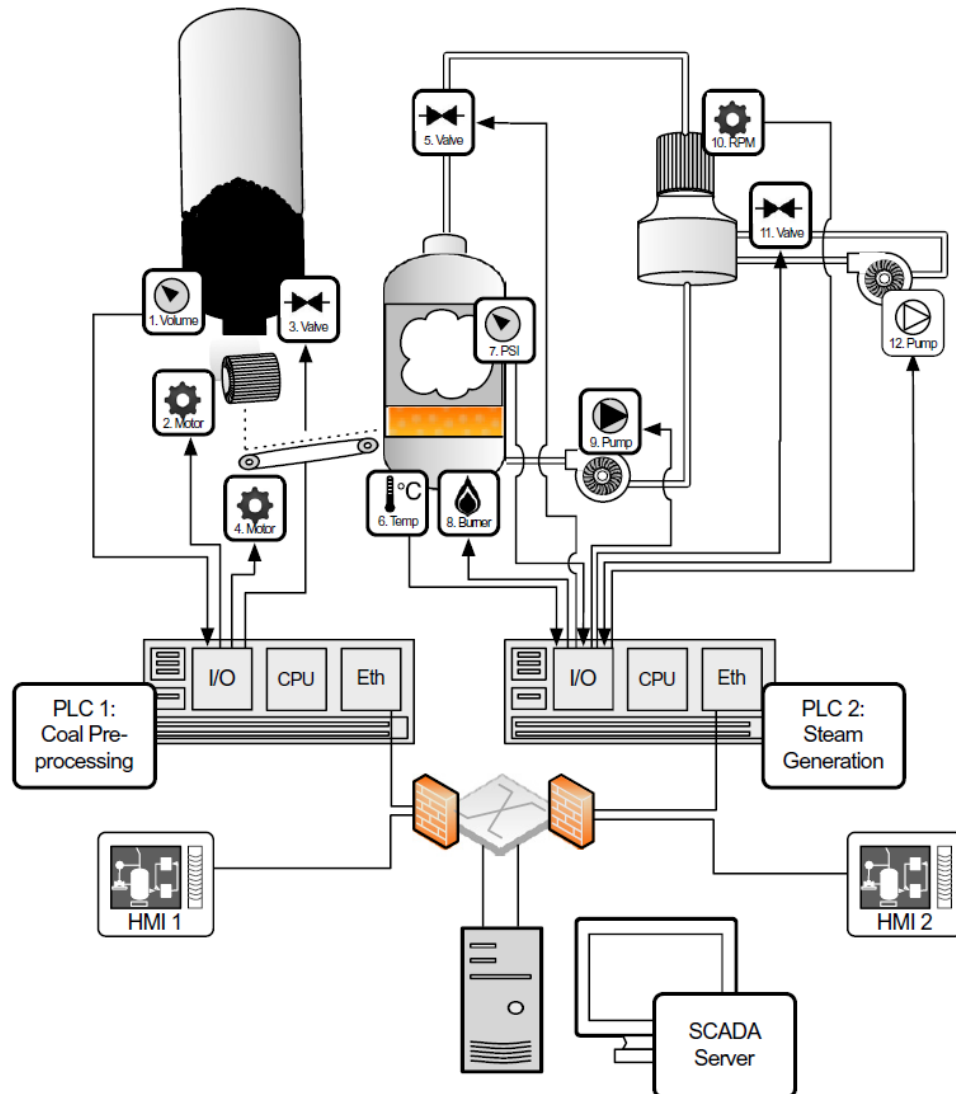
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IT-OT Communication Networks for Power Grids

SCADA for Generation and Transmission & Distribution Systems

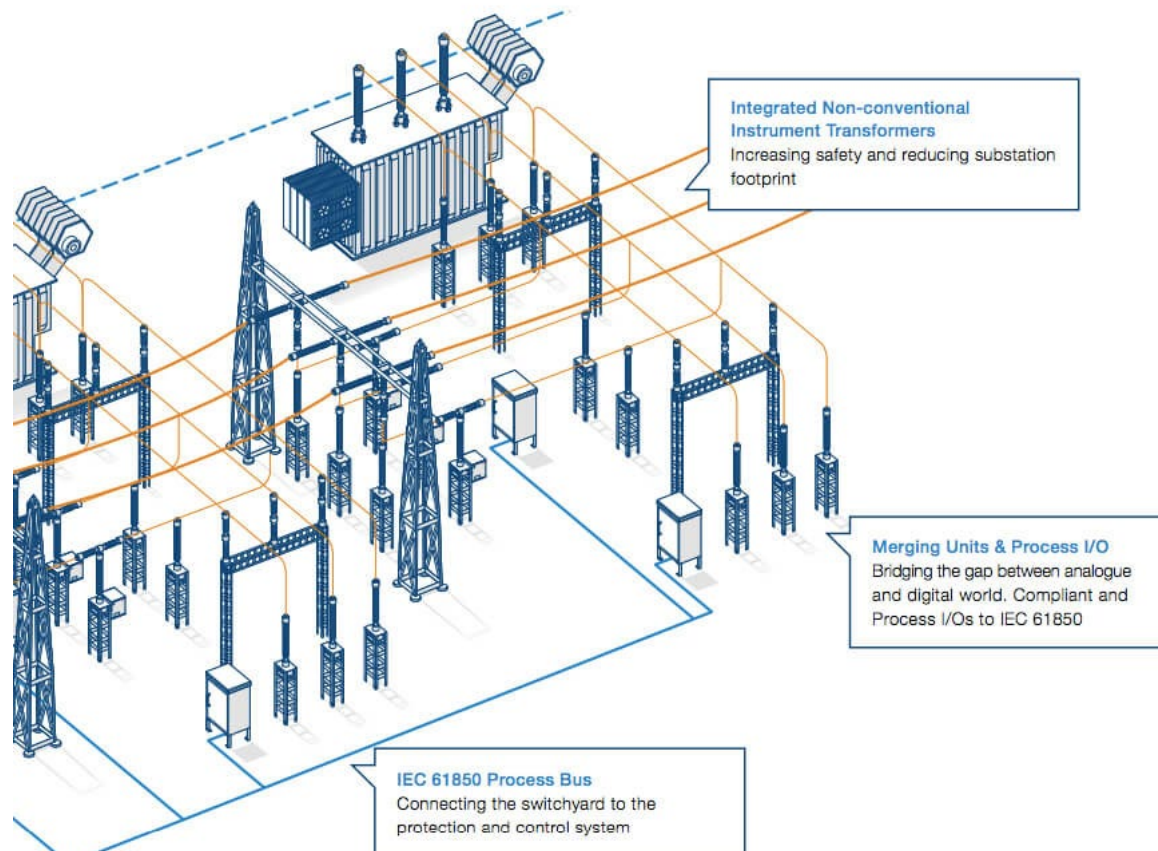


[Knapp, E. *et. al.* (2013). Applied cyber security and the smart grid: implementing security controls into the modern power infrastructure. Syngress]

Information Technology (IT) – Operational Technology (OT) Systems for Power Grids: Substations

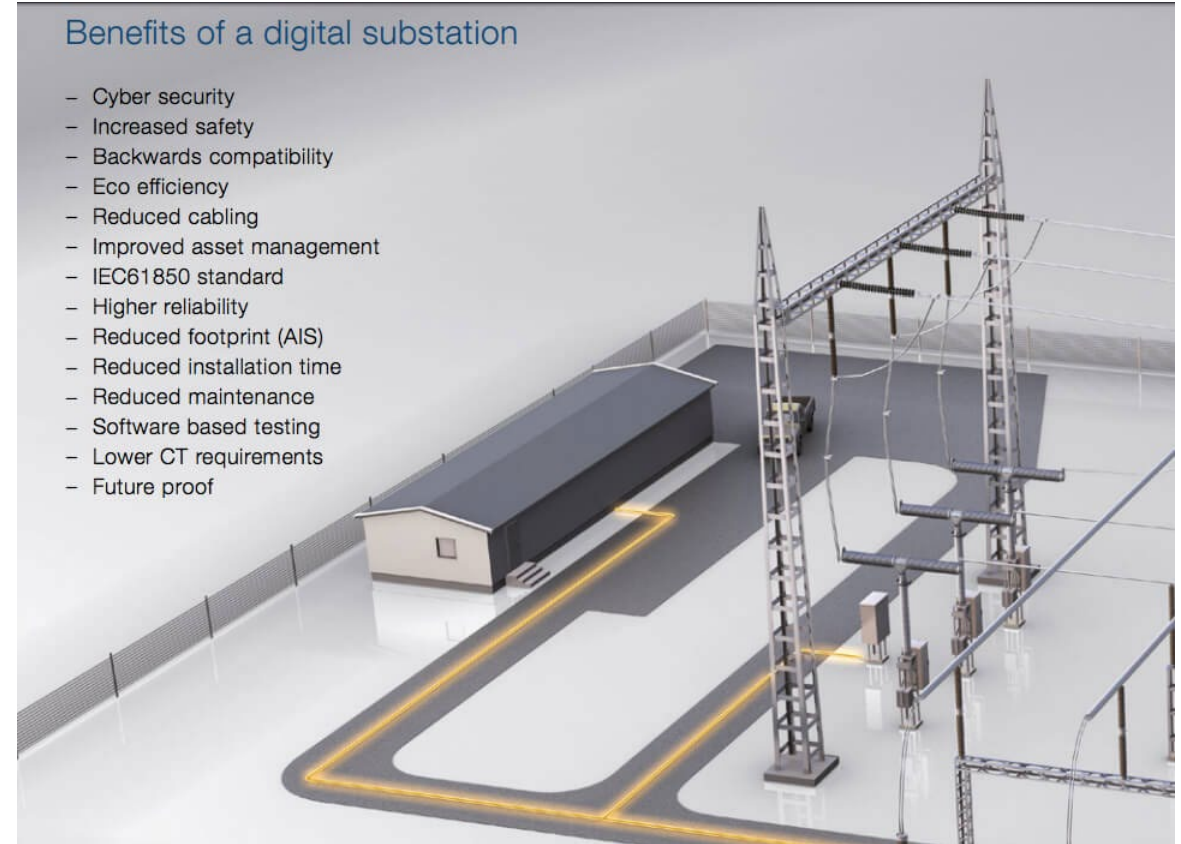


IT-OT Systems for Power Grids: Substation Monitoring and Control

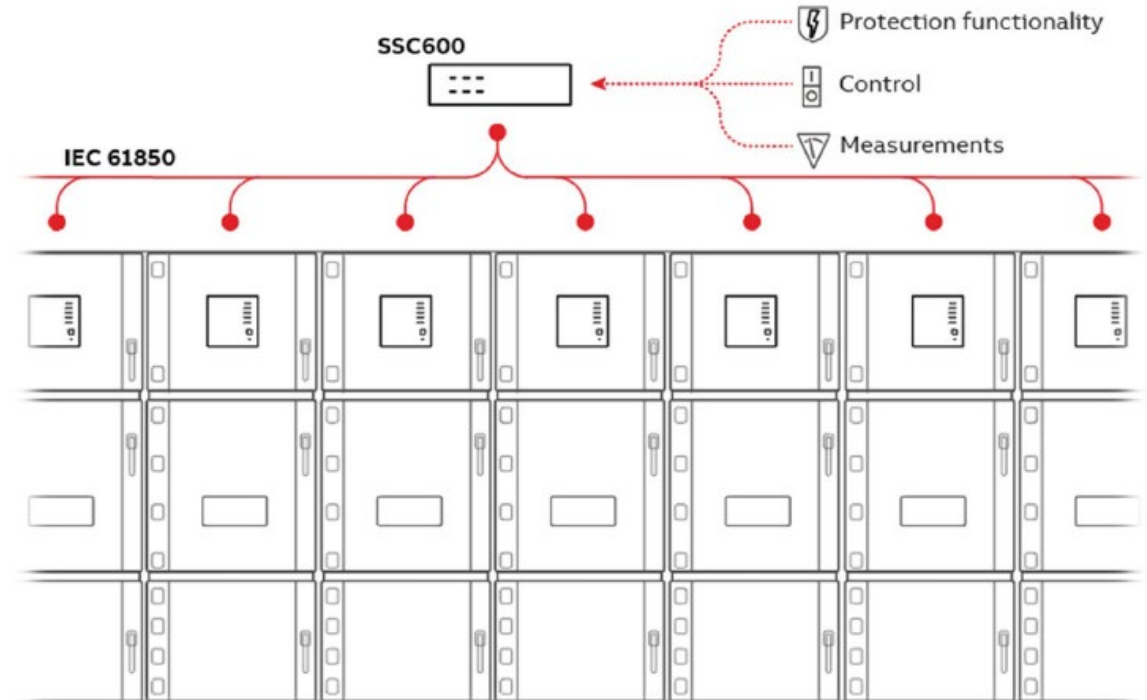
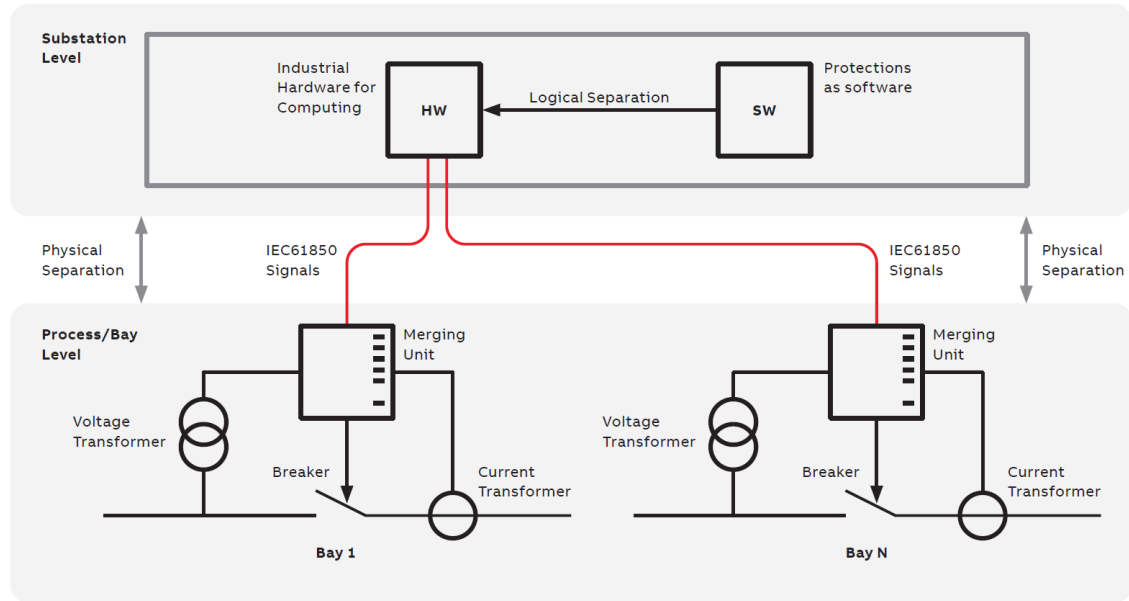


Benefits of a digital substation

- Cyber security
- Increased safety
- Backwards compatibility
- Eco efficiency
- Reduced cabling
- Improved asset management
- IEC61850 standard
- Higher reliability
- Reduced footprint (AIS)
- Reduced installation time
- Reduced maintenance
- Software based testing
- Lower CT requirements
- Future proof

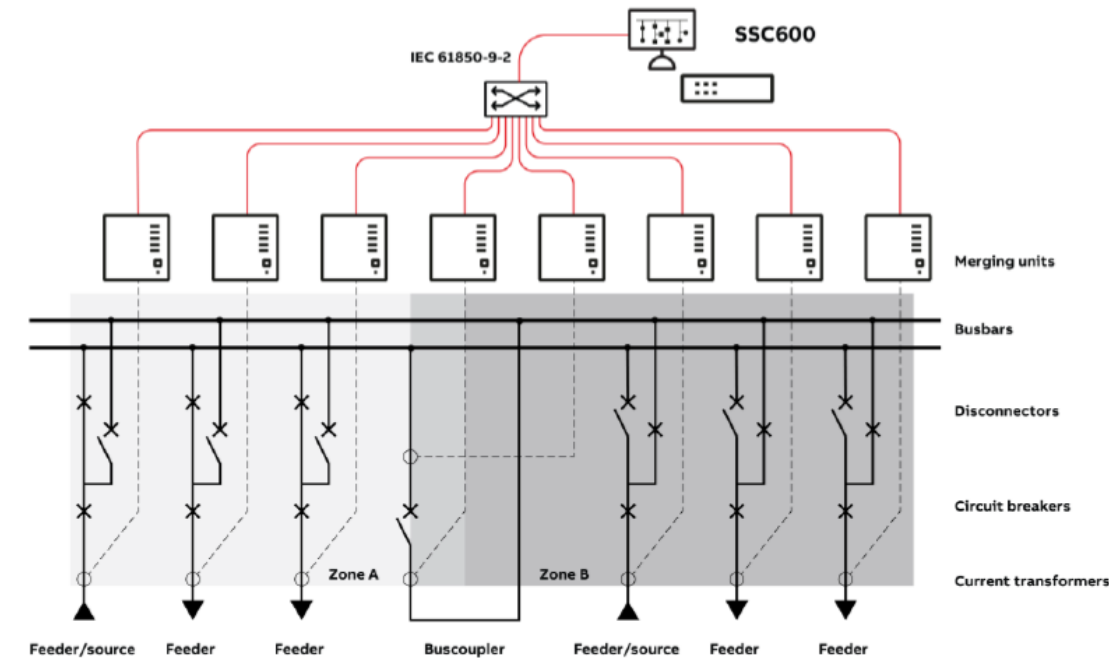
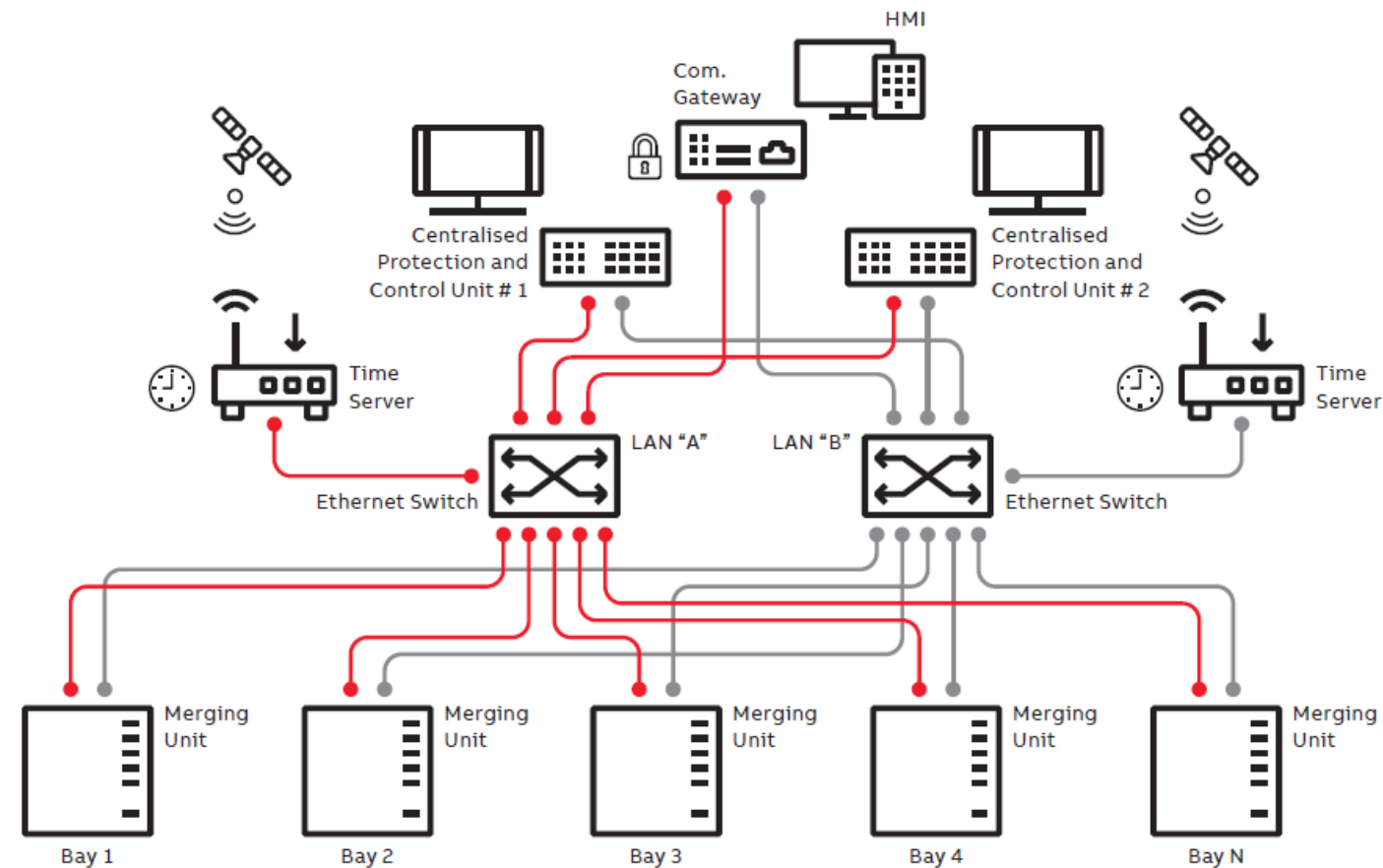


IT-OT Systems for Power Grids: Substation Monitoring and Control



Source: ABB <https://new.abb.com/medium-voltage/distribution-automation/campaigns/smart-substation-control-and-protection-ssc600>

IT-OT Systems for Power Grids: Substation Monitoring and Control



Source: ABB Whitepaper - Centralized Protection and Control.
<https://new.abb.com/medium-voltage/distribution-automation/campaigns/smart-substation-control-and-protection-ssc600>

IT-OT Systems for Power Grids: Substation Monitoring and Control

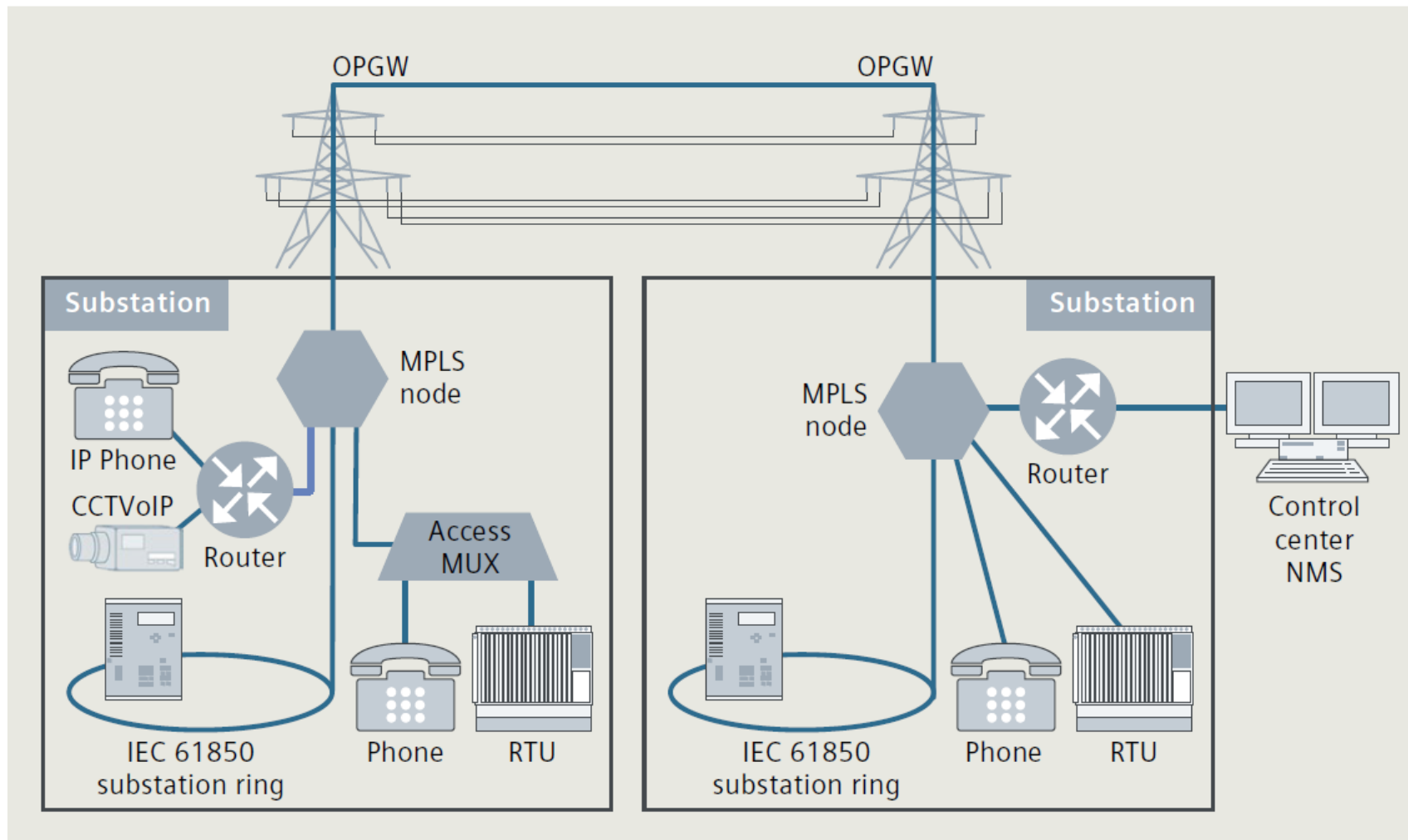


Source: <https://www.eq-house.com/>



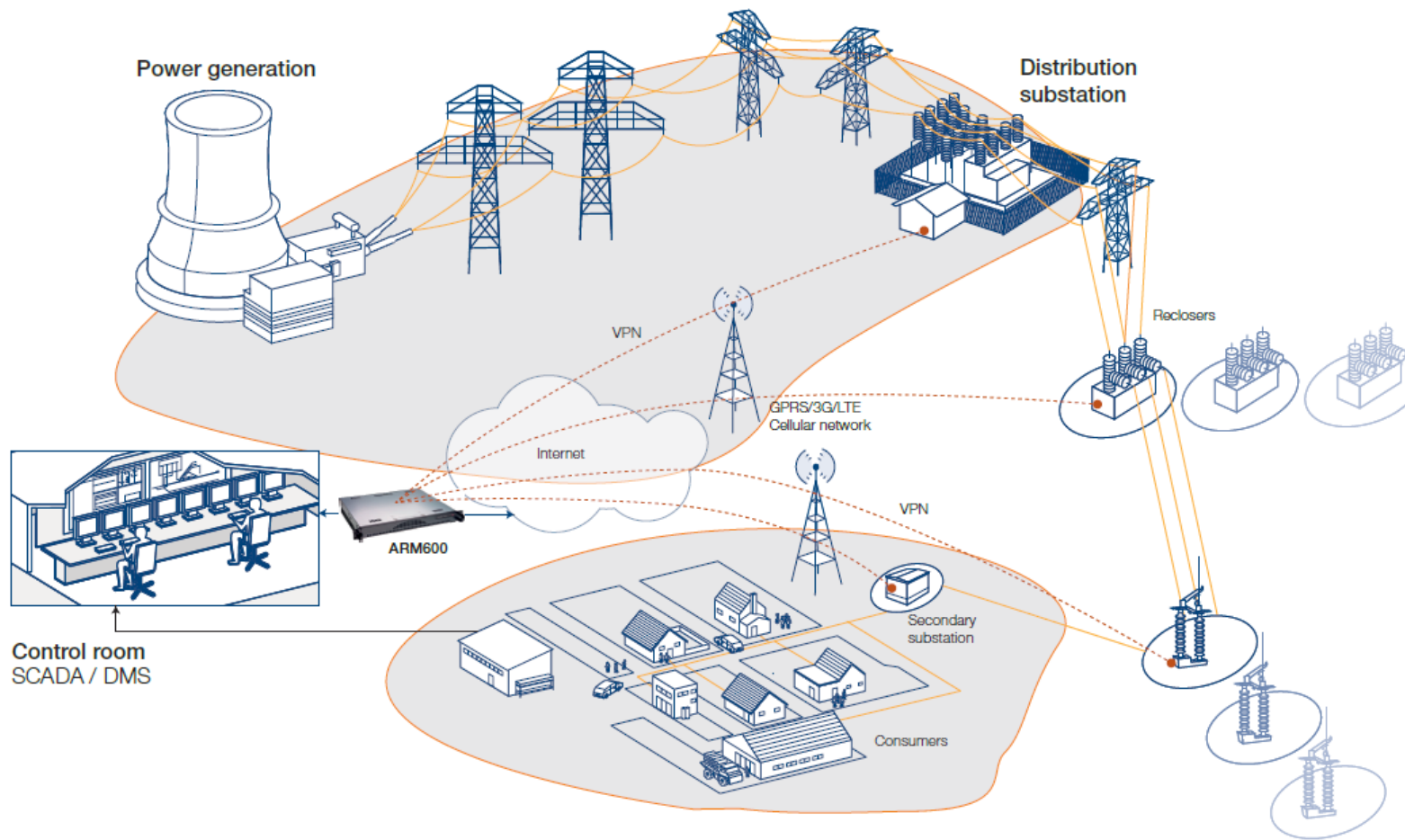
Source:
<https://www.windpowerengineering.com/going-digital-look-modern-substation/>

IT-OT Systems for Power Grids: Wide Area Communication



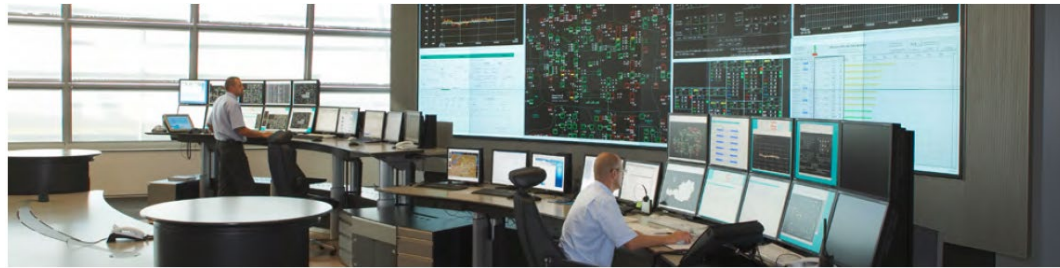
Source: Siemens Power Engineering Guide 8
<https://new.siemens.com/global/en/products/energy/topics/power-engineering-guide.html>

IT-OT Systems for Power Grids: Wide Area Communication

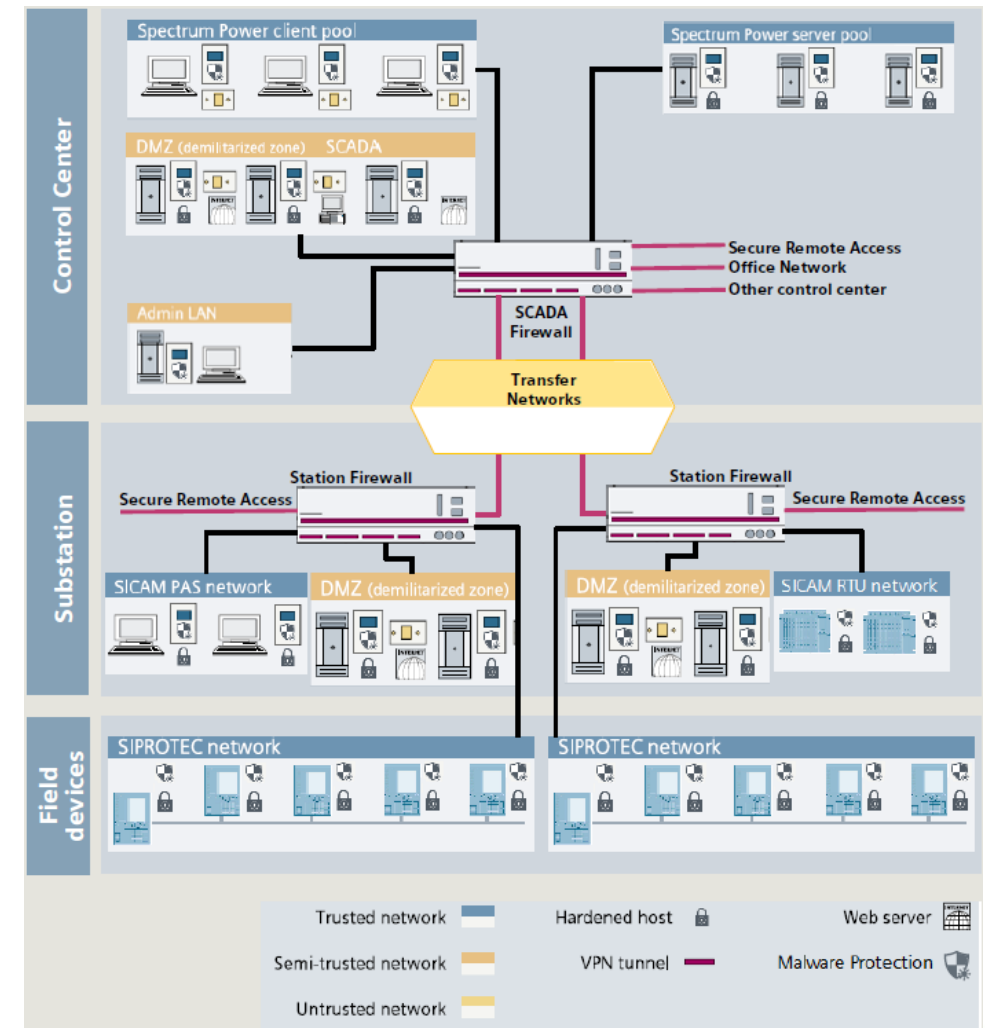


Source: <https://new.abb.com/medium-voltage/digital-substations/communication-devices/arctic-family/m2m-gateway-arm600>

IT-OT Systems for Power Grids: National Control Centres

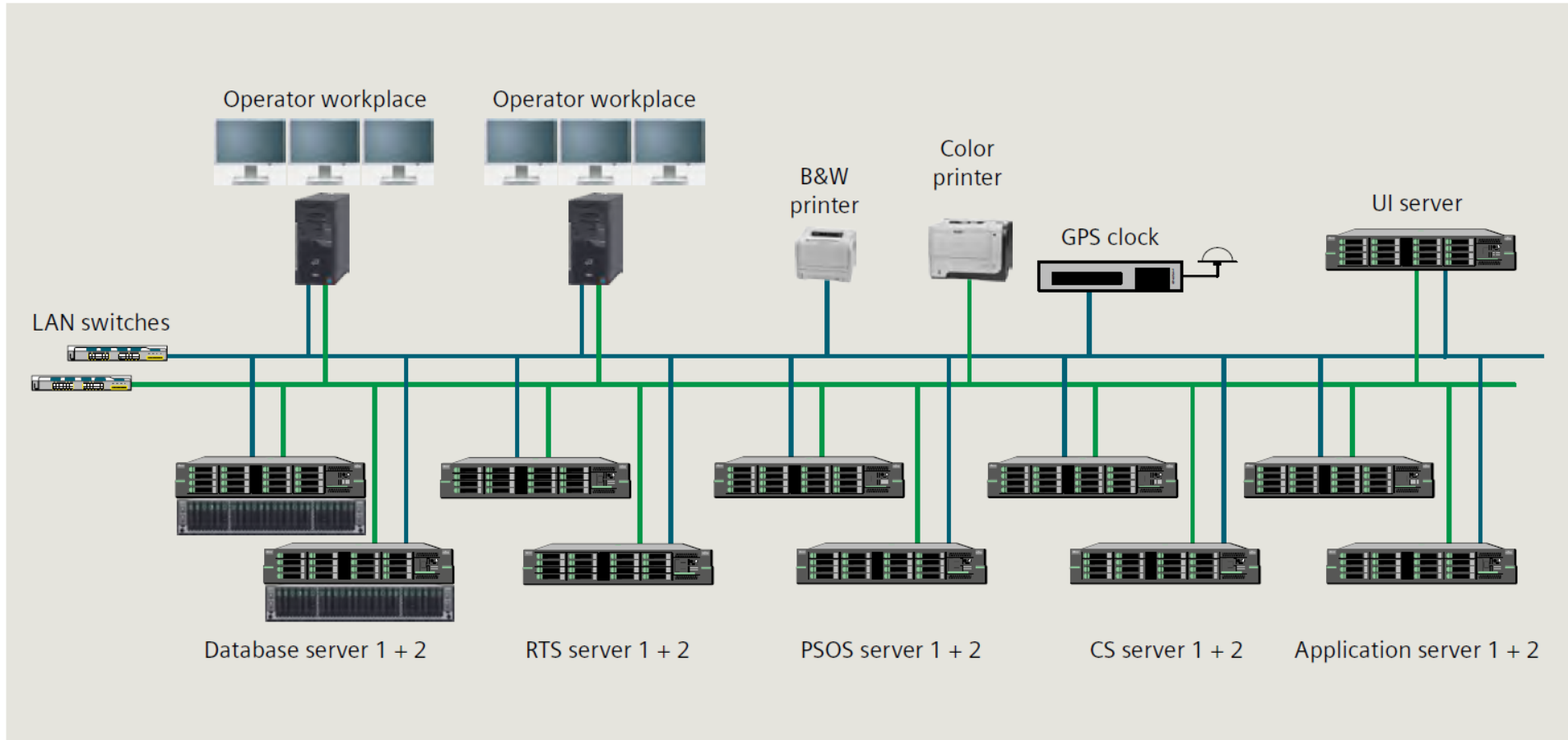


Source: ESB Networks 2027
www.esbnetworks.ie

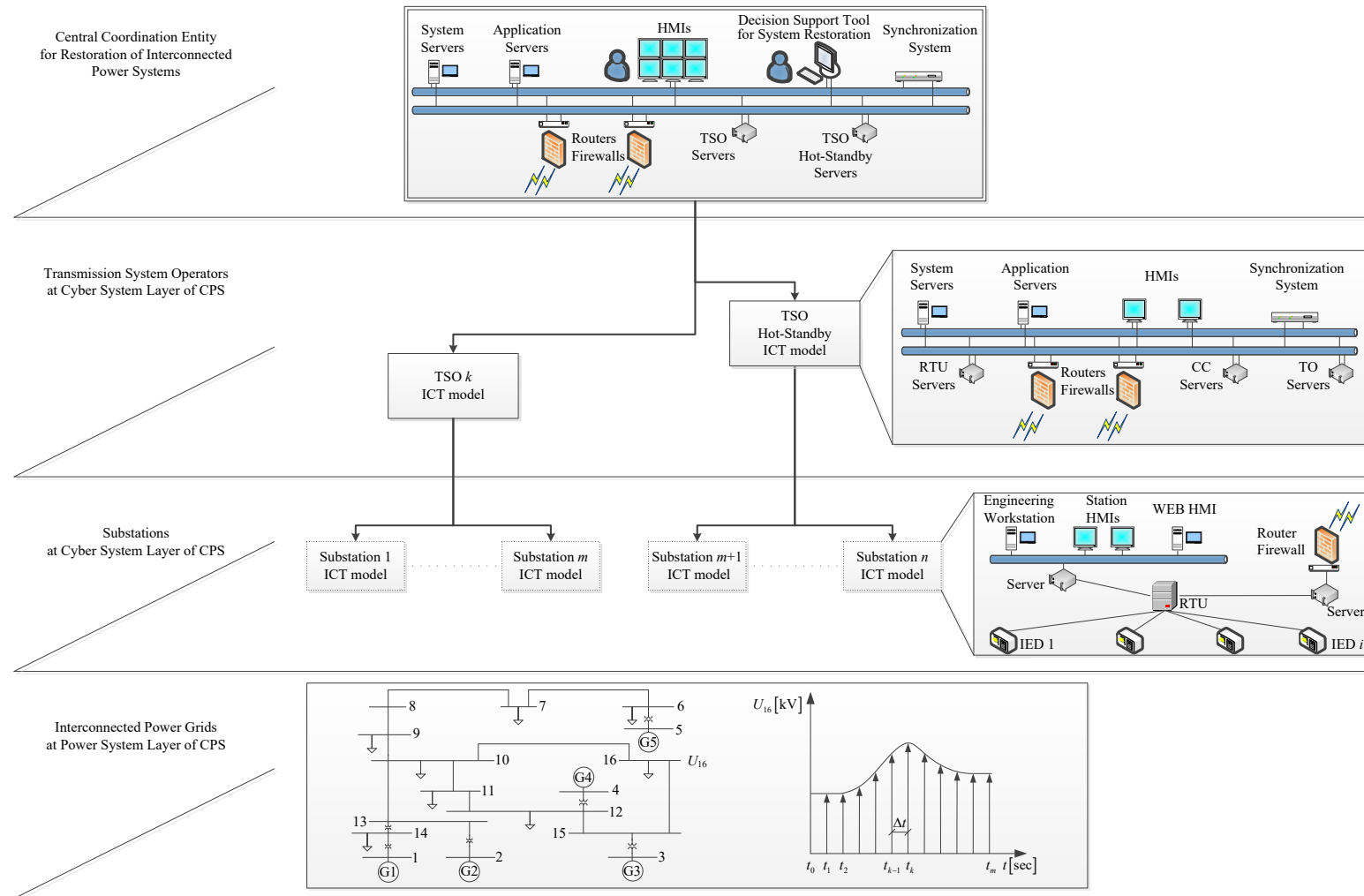


Source: Siemens Power Engineering Guide 8
<https://assets.new.siemens.com/siemens/assets/api/uuid:5bfb815b0db95760272f17c1329cc56c0c402686/peg8-final-160812.pdf>

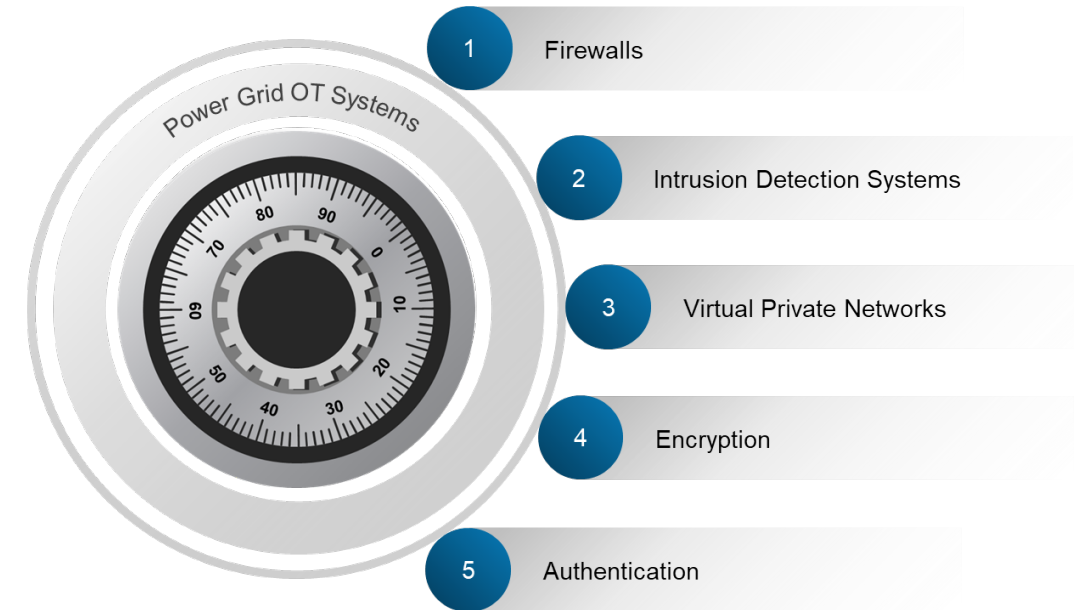
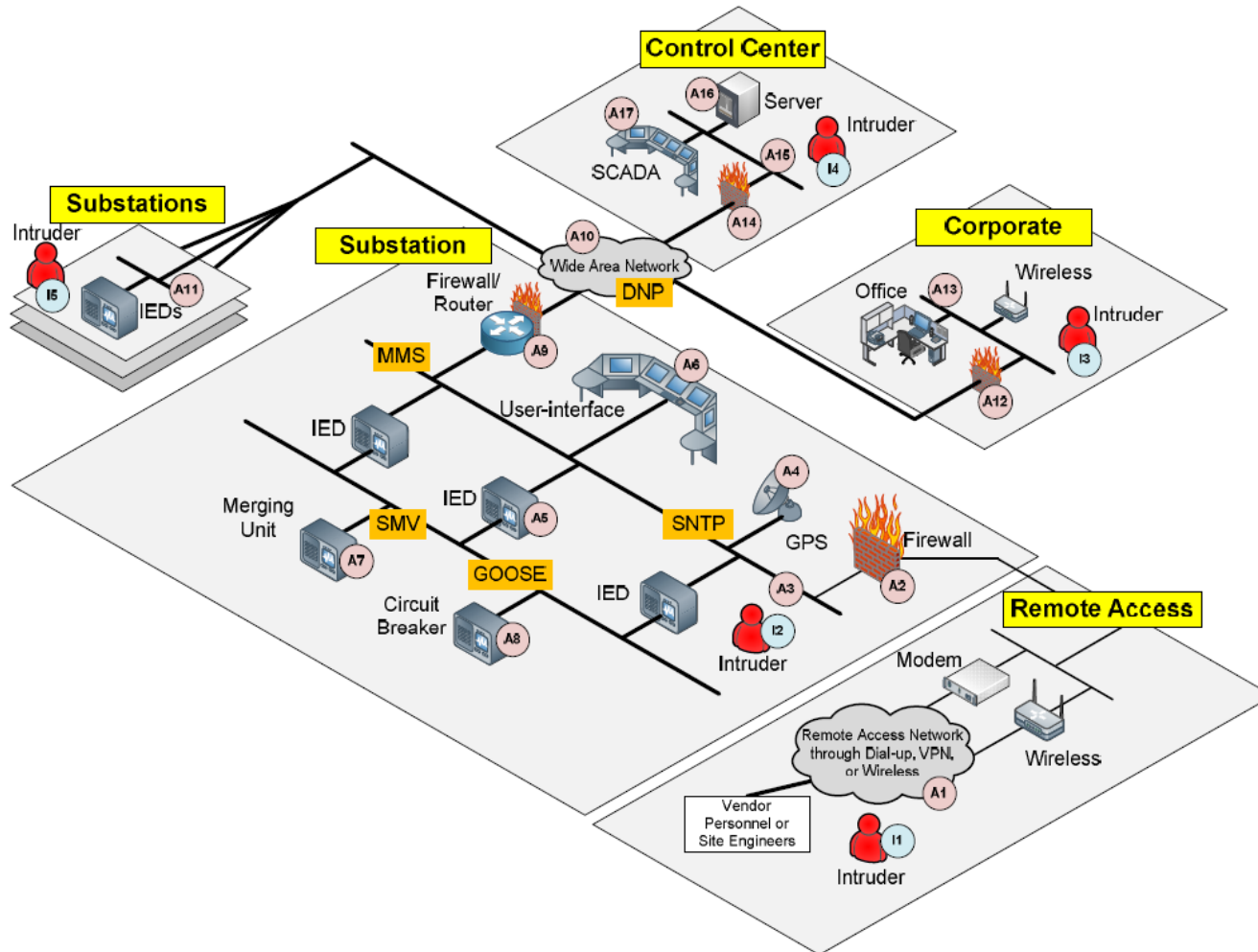
IT-OT Systems for Power Grids: National Control Centres



Cyber-Physical Systems (CPS)



IT-OT Systems for Power Grids and Cyber Security



[Stefanov *et al.* Probabilistic Reliability Analysis of Power Systems, Springer, 2020]

[Hong *et al.* Cyber Physical Systems Approach to Smart Electric Power Grid, Springer, 2015]



SCADA Vulnerabilities, Threats and Attack Vectors

Threats Affecting Power Grids

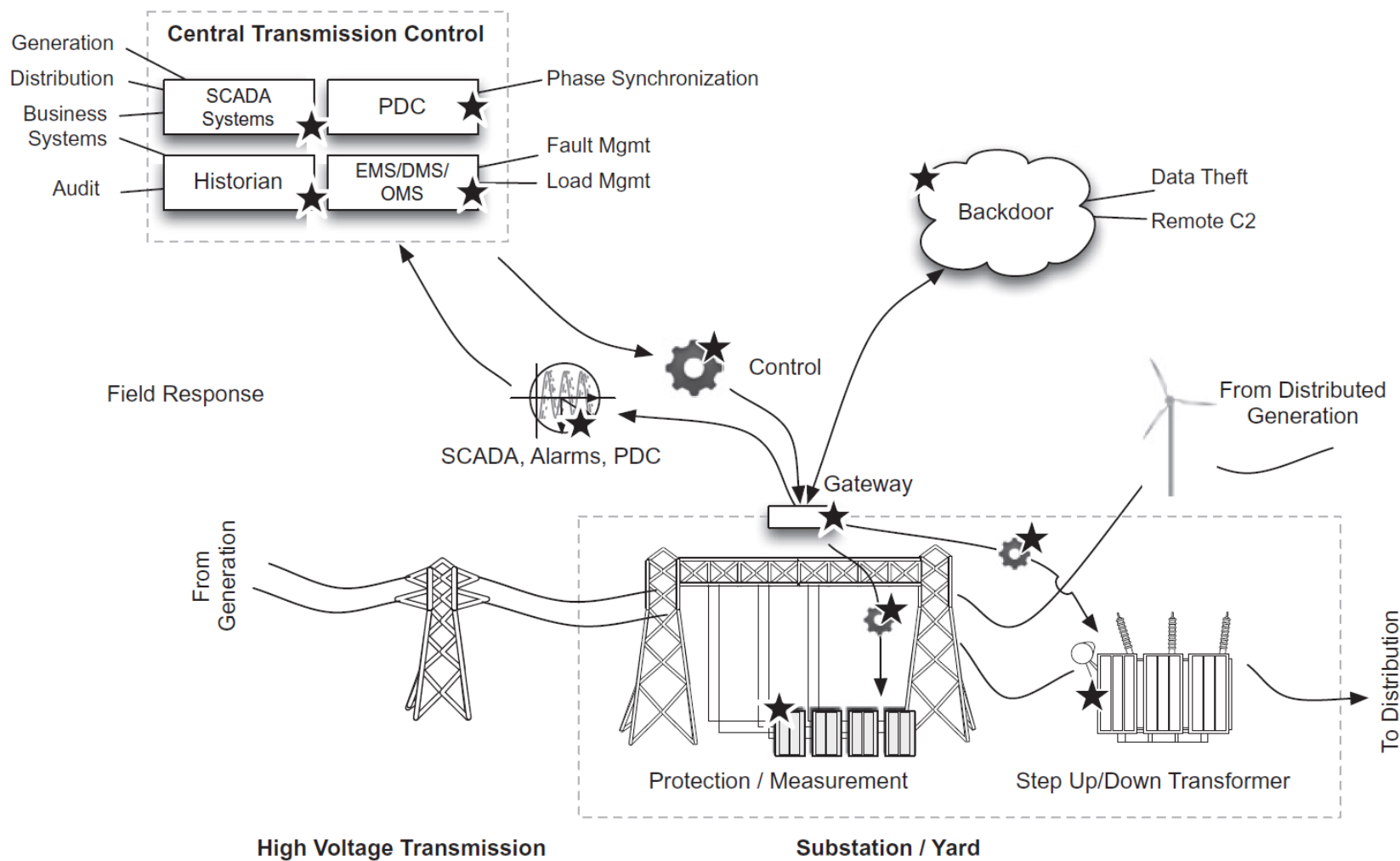


Likelihood: Low Medium Very high

Impact: Medium /High High High/Crucial Crucial

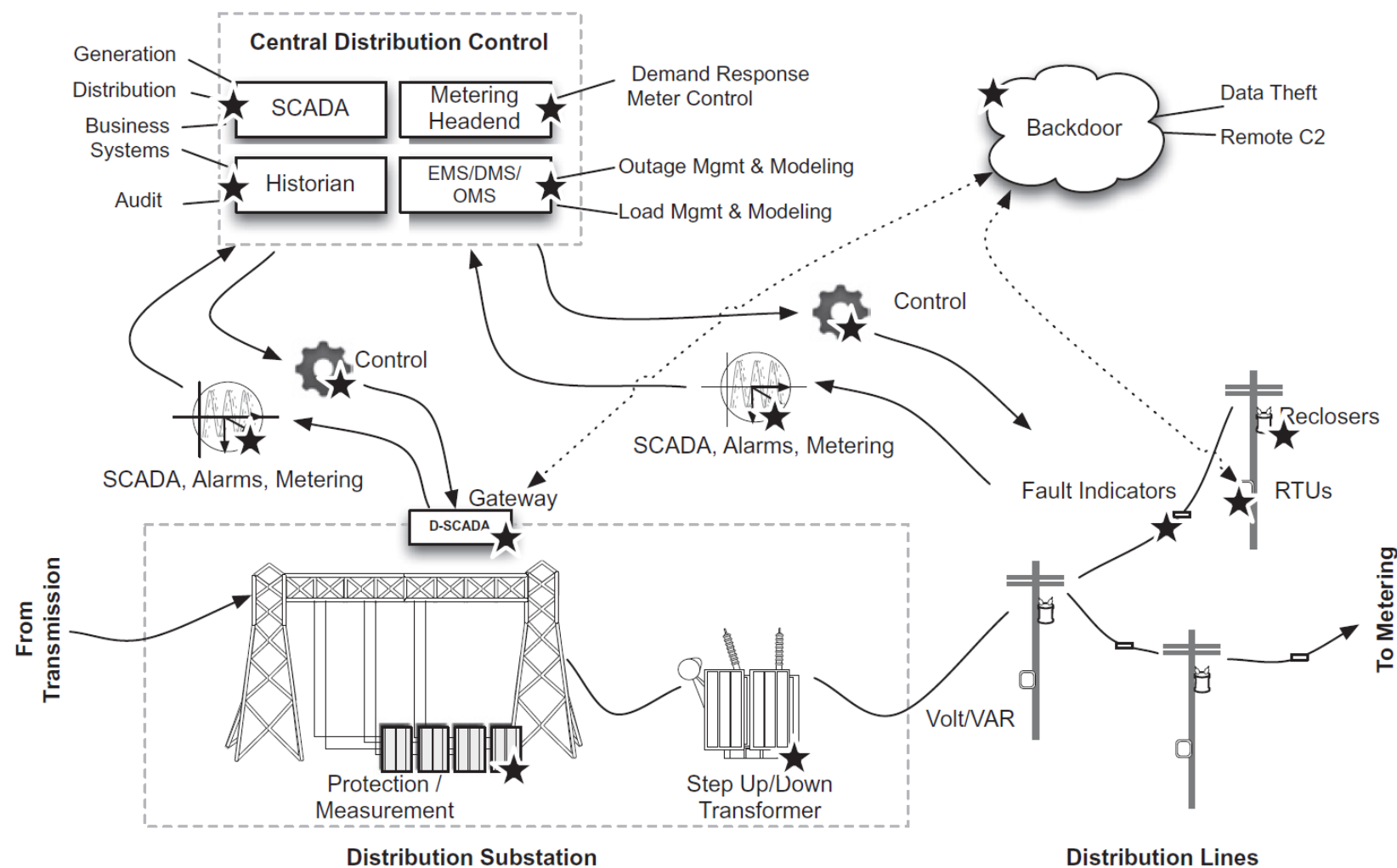
[ENISA. (2017). Communication network dependencies for ICS/SCADA systems, ISBN: 978-92-9204-192-2, doi: 10.2824/397676]

Transmission System Threats



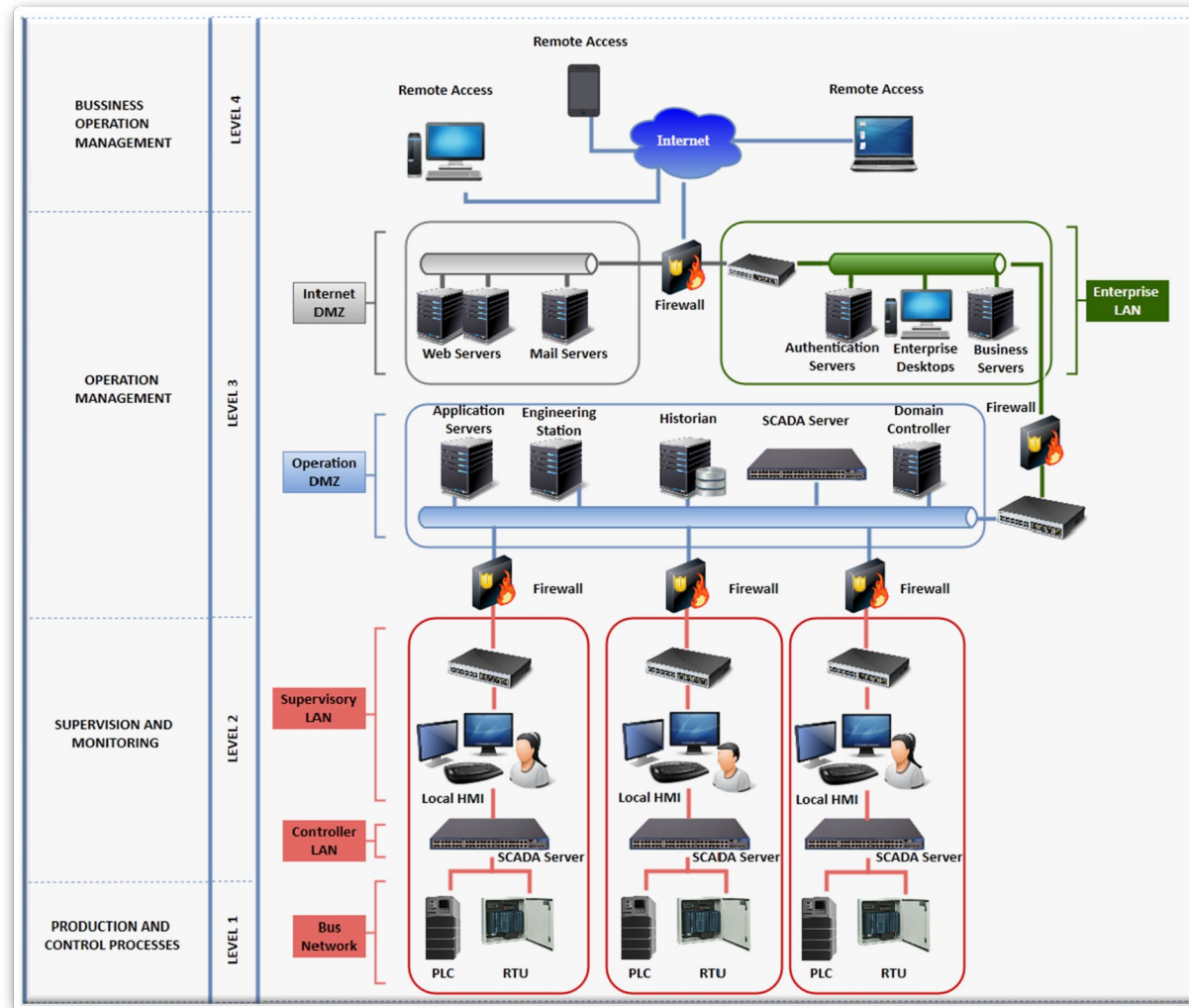
[Knapp, E. *et. al.* (2013). Applied cyber security and the smart grid: implementing security controls into the modern power infrastructure. Syngress]

Distribution System Threats



[Knapp, E. *et. al.* (2013). Applied cyber security and the smart grid: implementing security controls into the modern power infrastructure. Syngress]

SCADA System Threats





Source: thesoftwarereport.com

- Common vulnerabilities of SCADA systems

- **Non-existent monitoring process**

- ✓ Without active network monitoring, it is very difficult to detect suspicious activity, identify potential threats, and quickly react to cyber attacks
- ✓ The use of Intrusion Detection Systems (IDS) is not as common as in IT networks
- ✓ Even if IDS are in-place they may not be able to fully understand SCADA protocols
- ✓ This can be partially addressed by implementing anomaly detection systems
- ✓ Firewalls and antivirus are more common, but it is not a universal solution and does not cover all the risks

- **Deficient traffic content understanding**

- ✓ Managers need to know what type of traffic is going through their networks
- ✓ Needed to make informed decisions on how to respond to potential threats and on which kinds of traffic to allow and which to filter
- ✓ This also helps to establish proper segregation and network segmentation



Source: thesoftwarereport.com

- Common vulnerabilities of SCADA systems

- **Staff inexperienced in cyber security related topics**

- ✓ SCADA system staff and operators are familiar with keeping control systems running
- ✓ The normal goals of reliability and availability can initially feel in conflict with security efforts
- ✓ The important role of developing security policies can be a foreign concept to typical SCADA staff, who typically focus on engineering and technical solutions to problems
- ✓ SCADA staff may not be receptive to IT staff recommendations

- **Operating system vulnerabilities**

- ✓ All normal IT operating system vulnerabilities are present in SCADA systems
- ✓ The difference from an IT system is that patching may be performed less rigorously
- ✓ It is usual for a SCADA system operator to have a running system that is expected to perform without interruptions

- **Slow / lack of updates**

- ✓ Maintaining SCADA firmware and software up-to-date is not easy, and it can be very complex for critical infrastructure systems
- ✓ An update error could cause severe issues on the whole system
- ✓ Cyber fragility results from applying a change to the system without having tested it beforehand and having foreseen its effects



Source: thesoftwarereport.com

- Common vulnerabilities of SCADA systems

- Remote processor operations

- ✓ Certain classes of remote processors have known security vulnerabilities. The difficulties are
- ✓ The computation power and memory resources of the processors are modest and not suitable for security upgrades
- ✓ Once they are installed, they typically stay in place for ten years or more
- ✓ The result is vulnerable equipment that stays vulnerable for a long time

- SCADA software features

- ✓ SCADA software usually provides basic and modest security features
- ✓ These are not always enabled by default and could act as additional weaknesses if operators are unaware of the need of enabling these features

- Inappropriate applications installed on critical SCADA host computers

- ✓ Very few security measures are used in SCADA host computers
- ✓ This leads to an operator or administrator inadvertently installing an inappropriate application on a critical system network device



- Common vulnerabilities of SCADA systems

- **Lack of knowledge regarding the devices**

- ✓ Since most SCADA systems have been developed gradually over time, it's not uncommon to see technology that is a couple of years old working alongside an industrial network environment
- ✓ Knowledge transfer regarding functioning and maintenance of ageing devices should be ensured

- **Authentication weaknesses**

- ✓ Authentication solutions can easily be defeated if the solution is not properly implemented (allowing weak passwords, user credential sharing, no user logging)
- ✓ Or in the case of older devices, which make use of weaker, more primitive authentication methods
- ✓ Moving to two-factor authentication is limited by work conditions that may impede iris scans or fingerprint scans because of dirty hands or the wearing of safety goggles
- ✓ Confidentiality and authentication is often compromised by using clear text transmissions. This weakness eliminates authentication and accountability validity

- **Unauthenticated Remote Terminal Unit (RTU) network connections**

- ✓ Older SCADA systems lack basic security features, so it is imperative for organizations that own such systems to insist on vendors to provide security measures in the form of product patches and upgrades which can protect the system from unauthenticated RTU network connections



Source: thesoftwarereport.com

- Common vulnerabilities of SCADA systems

- Remote access supervision

- ✓ Because of economics for staffing control centres around the clock, it is not uncommon for SCADA systems to be configured with remote access
- ✓ This can include Virtual Private Network (VPN) access over the Internet. These scenarios should be controlled and monitored, and should include, at least, the same security measures as internal connections

- Interconnection management

- ✓ The more connections, the more exposure a SCADA system has
- ✓ Economic and enterprise pressures often result in the existence of internal connections between the SCADA network and the business network in order to allow remote access, control and/or maintenance

- Wireless connections

- ✓ SCADA systems often use microwave, data radios and cellular packet services for communications
- ✓ Depending on the implementation, these forms of communication can be vulnerable to various types of attacks

Threats, Vulnerabilities and Attack Vectors



Source: thesoftwarereport.com

- Common vulnerabilities of SCADA systems

- Available public information

- ✓ In the past it was not unusual for SCADA system owners to publish information on the design of their systems, as security was not a top concern and most devices were not interconnected
- ✓ It is also common for consultants/contractors to advertise past experience including information regarding the systems they have worked on
- ✓ Both scenarios can result in the exposure of system vulnerabilities which is more serious as these systems have now become interconnected

- The wrong belief that SCADA systems have the benefit of security through obscurity

- ✓ The use of closed-source proprietary protocols does not provide security, and it can be counterproductive
- ✓ Security by obscurity is not a good practice and usually gives users a false sense of security, while in fact they are at greater risk

- The wrong belief that SCADA systems are isolated

- ✓ Just because a SCADA network is not connected to the Internet does not make it secure
- ✓ For maintenance and support reasons devices belonging to segmented networks are sometimes exposed to Internet through VPNs or remote access connections
- ✓ These connections should be controlled and be enabled on-demand only when required

Threats, Vulnerabilities and Attack Vectors



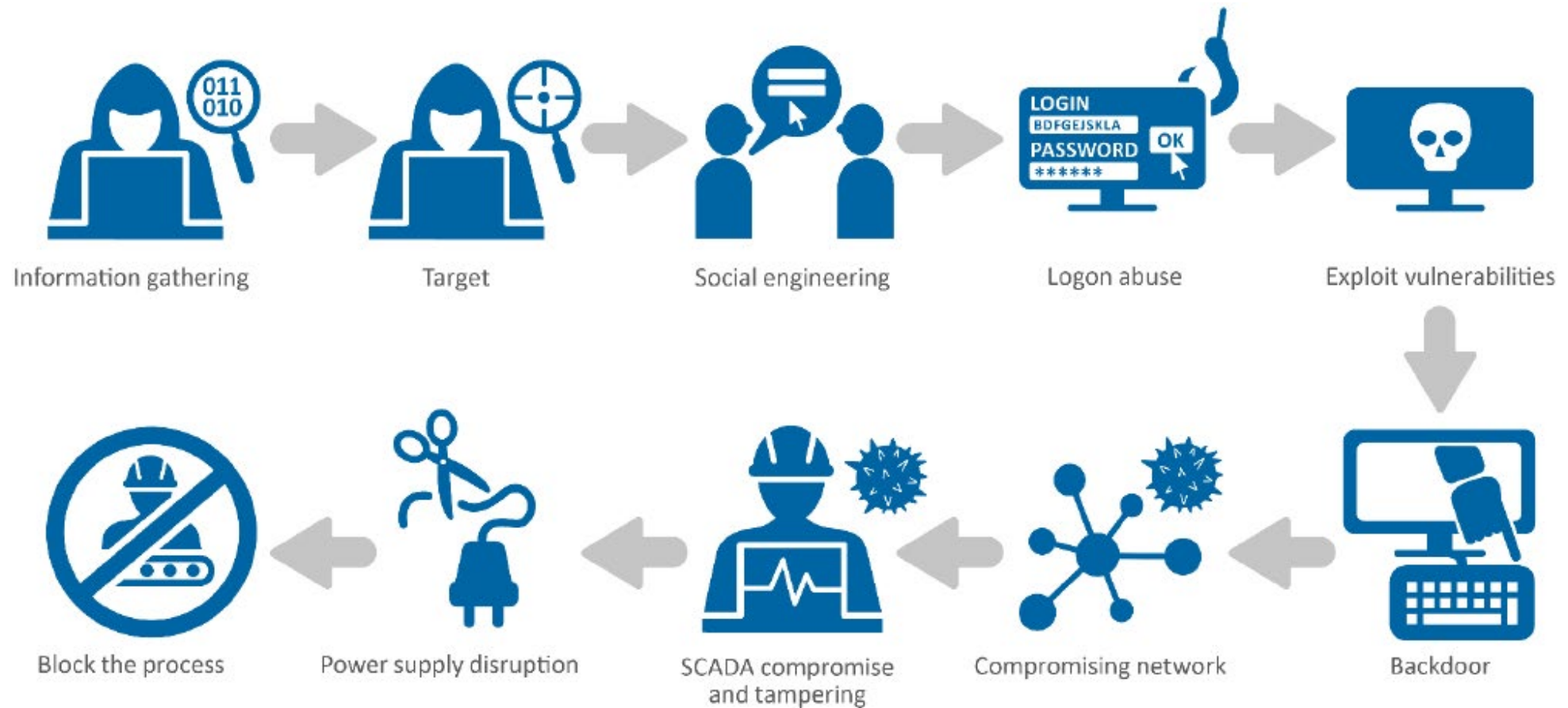
Source: thesoftwarereport.com

- Attack vectors on SCADA systems

SAMPLE ATTACK SCENARIOS	IMPORTANCE LEVEL
1. Against the administration systems of SCADA	Crucial
2. Against actuators	High / Crucial
3. Against the network link between sensors/actuators and HMI or controller	High
4. Against sensors	Medium / Crucial
5. Against the information transiting the network	Medium / Crucial
6. Compromised ICT components as backdoors	Medium / Crucial
7. Exploit Protocol vulnerabilities	Medium / High
8. Against Control Data Historians, Local HMIs or controllers	Medium

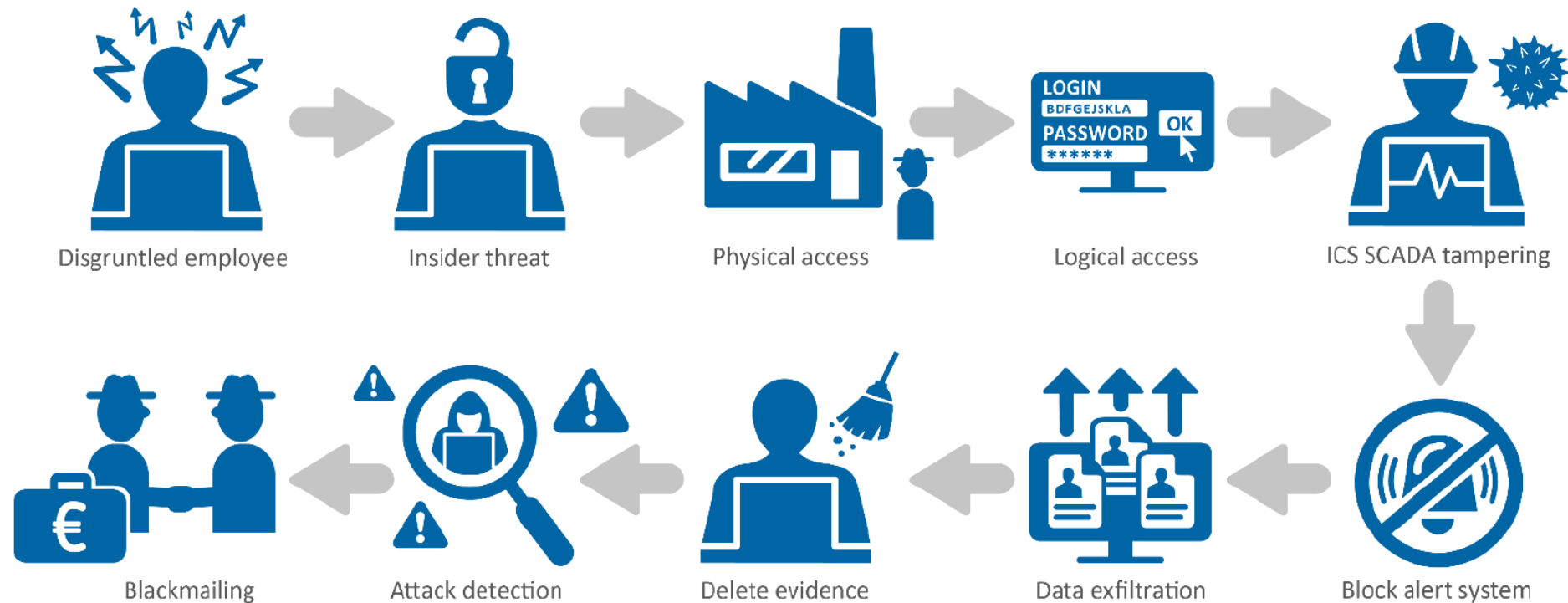
Hacking IT-OT Systems of Power Grids

- Attack scenario 1: SCADA system compromise



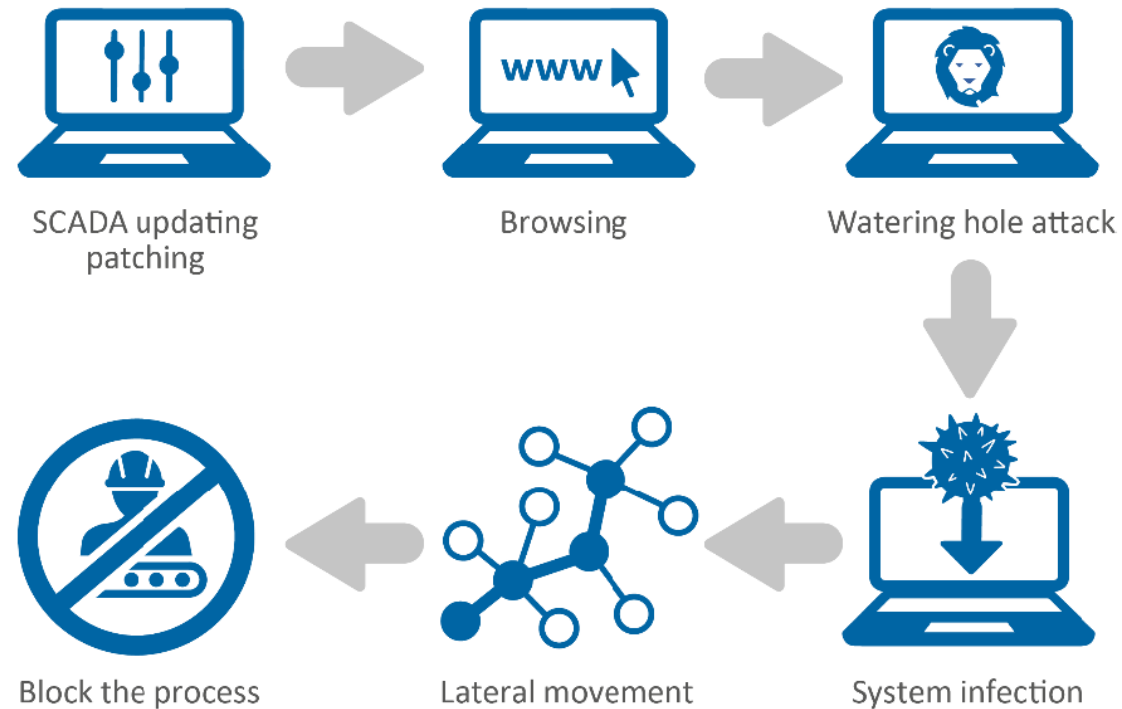
Hacking IT-OT Systems of Power Grids

- Attack scenario 2: Insider threat



Hacking IT-OT Systems of Power Grids

- Attack scenario 3: Malware infection



Are Cyber Attacks a Real Threat?

“Aurora” project (March 2007)

- Demo of targeted cyber attack
- US Department of Energy’s Idaho National Laboratory
- Targeted the control system of an electrical generator (Diesel 2.25 MW)
- Generator out of control and damaged

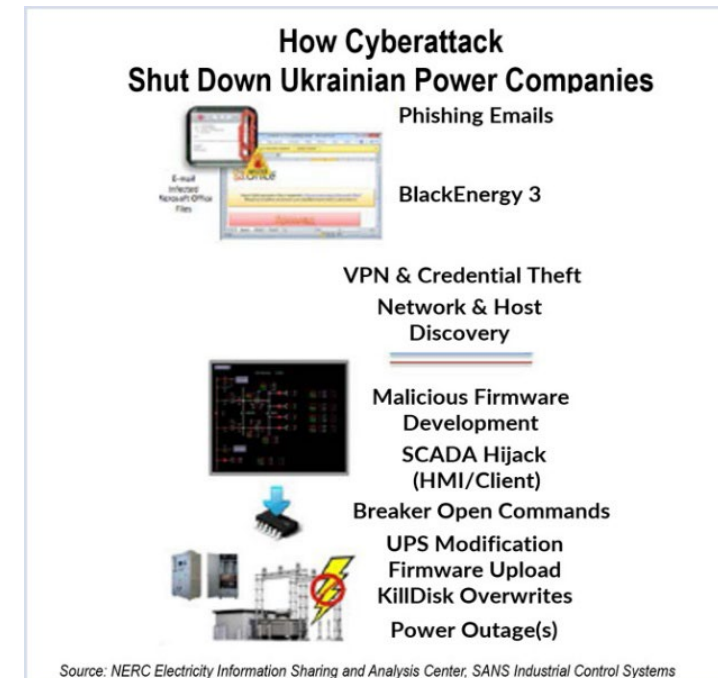


Source: <https://edition.cnn.com/>

Are Cyber Attacks a Real Threat?

Cyber attack on the power grid in Ukraine (December 23, 2015)

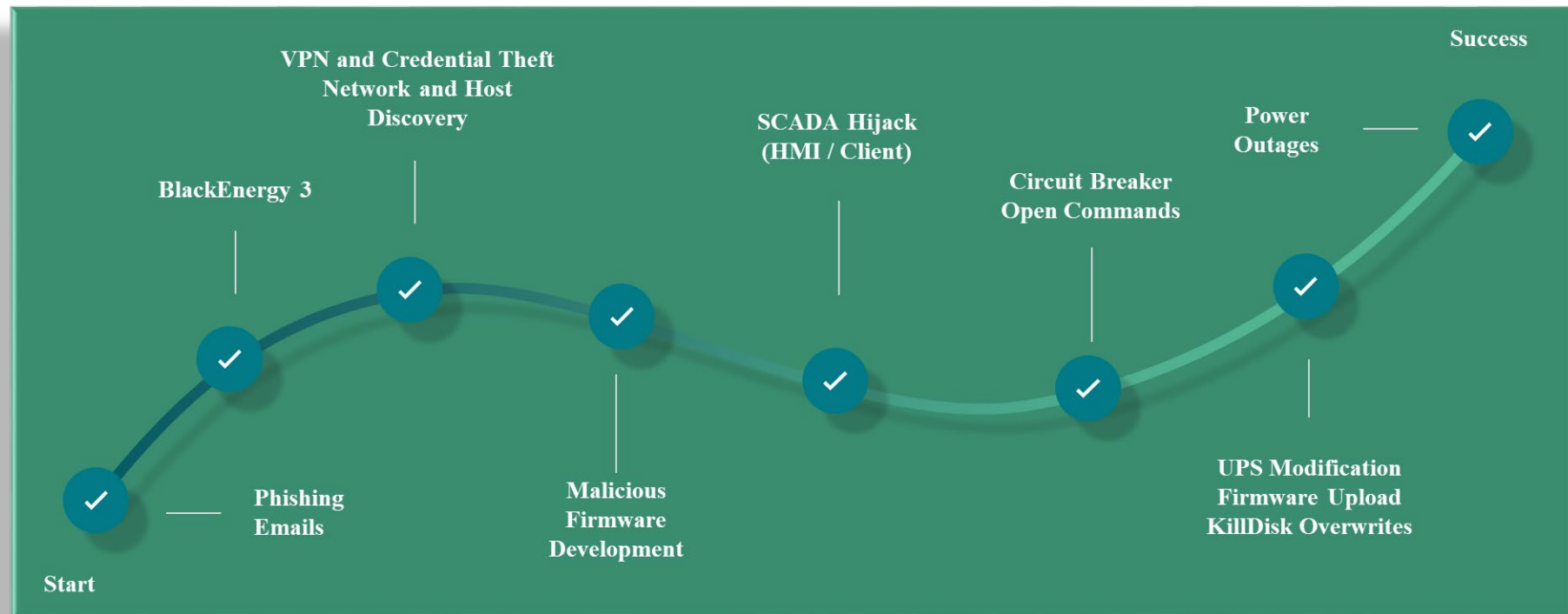
- Attackers intruded into IT and SCADA of three DSOs
- Seven 110 kV and twenty three 35 kV substations disconnected from the grid for 3 hours
- Cyber attack resulted in power outages that affected 225,000 customers



Are Cyber Attacks a Real Threat?

Cyber attack kill chain in Ukraine (December 23, 2015)

- Stage one: intrusion in IT and OT systems ~ 6 months
- Stage two: attack execution



Are Cyber Attacks a Real Threat?

Cyber attack on the power grid in Ukraine (December 17, 2016)

- SCADA at transmission level (targeted single 330 kV substation)
- Power outage in distribution system (200 MW load unsupplied)
- More sophisticated than 2015

	2015	2016
 Phishing	✓	✓
 OT accessed via IT system	✓	✓
 Remote control of SCADA	✓	✗
 Malware Industroyer	✗	✓



Cyber Security

Cyber Security: Definition

- Cyber security is the collection of defensive technologies (hardware and software), processes and practices
 - Designed to protect networks, computers, programs and information from
 - ✓ Attack
 - ✓ Damage or
 - ✓ Unauthorised access
 - In order to secure computer systems (IT-OT systems)
- Cyber security protects against threats using defensive measures
 - Information assurance
 - Computer systems and applications hardening
 - Malware protection
 - Access control
 - Information infrastructure protection
 - Network security



Source: yokogawa.com

Cyber Security: Definition

- Six security goals:

- **Confidentiality:**

- ✓ Only the intended receiver or user can understand the contents of the message

- **Authentication:**

- ✓ The identity of the individuals involved can be confirmed

- **Accountability and non-repudiation:**

- ✓ Every individual who works with an information system should have specific responsibilities for information assurance
- ✓ Non-repudiation is the assurance that someone cannot deny the validity of something
- ✓ Non-repudiation makes it very difficult to successfully deny who sent / where a message came from as well as the authenticity and integrity of that message



Source: contact.co.uk

Cyber Security: Definition

- Six security goals:

- **Integrity:**

- ✓ The assurance that the information has not been altered by elements along the communication path
 - Integrity for routing and the DNS infrastructure

- **Access control:**

- ✓ A security measure that permits access to various resources for legitimate users / processes

- **Availability:**

- ✓ Services must be accessible and available to allowed users / processes in an information infrastructure



Source: contact.co.uk

Cyber Security: Definition

- Cyber attacks

- Are offensive, malicious attempts to steal, expose, alter, disable or destroy information through unauthorised access to computer systems

- Network-based exploits

- Attacks to Operating System (OS), applications, hardware, and network equipment vulnerabilities
 - ✓ Malware
 - ✓ Configuration weaknesses
 - ✓ Syntax and semantics weaknesses
 - ✓ Validation weaknesses
- Attacks to confidentiality
 - ✓ Memory scraping
 - ✓ Eavesdropping
 - ✓ Packet sniffing
- Attacks on integrity
 - ✓ Modify content



Source: yokogawa.com

Cyber Security: Definition

- Network-based exploits

- Attack on Authenticity

- ✓ Identity theft
- ✓ Password crack
- ✓ Phishing attack
- ✓ DNS attack
- ✓ Cache poisoning

- Evasion on security equipment/measures

- ✓ Mutated attacks

- Attacks on Availability

- ✓ Distributed Denial of Service (DDoS)

- Social engineering



Source: yokogawa.com



Cyber Kill Chain and Security Controls

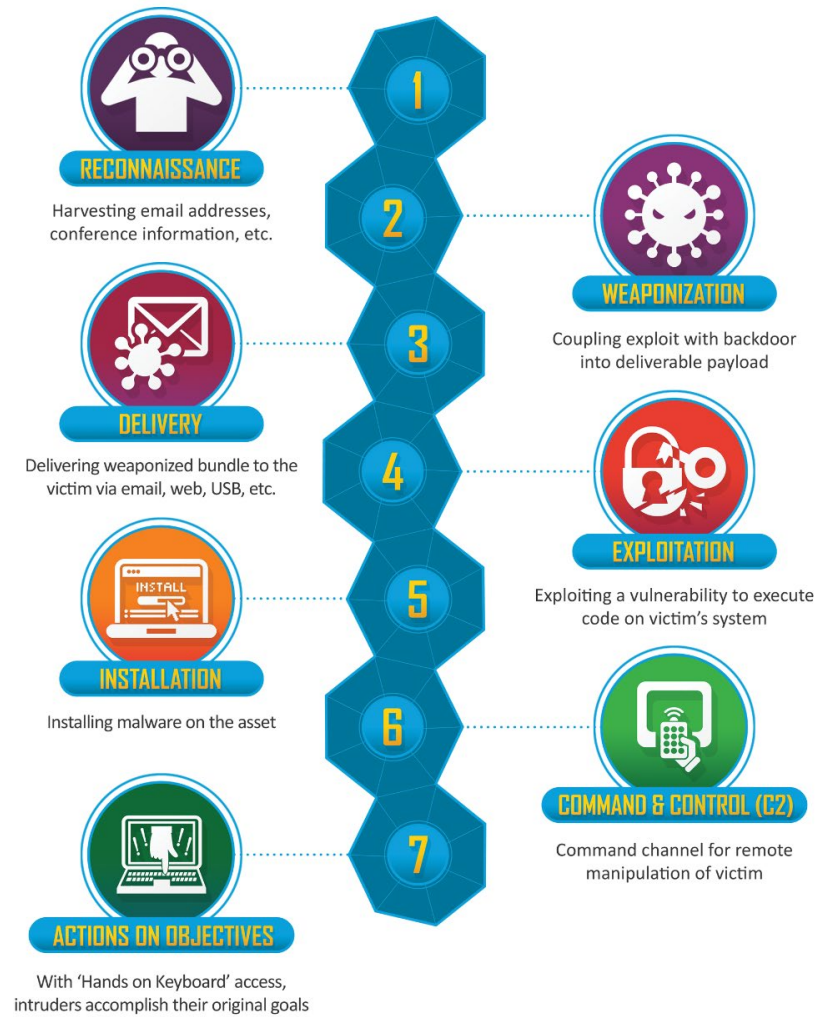
What Is the Cyber Kill Chain?

- What is a Cyber Kill Chain?
 - Cyber kill chain is a cyber security model created by Lockheed Martin used to:
 - Trace the stages of a cyber-attack, identify vulnerabilities and help security teams to stop the attacks at every stage of the chain
 - The term kill chain is adopted from military, which uses this term related to the structure of an attack
 - ✓ It consists of identifying a target, dispatch, decision, order, and finally, destruction of the target



Source: yokogawa.com

Cyber Kill Chain



What Is the Cyber Kill Chain?

How does the Cyber Kill Chain Work? The cyber kill chain consists of 7 distinct steps:

- Reconnaissance

- The attacker collects data about the target and the tactics for the attack. This includes harvesting email addresses and gathering other information
- Automated scanners are used by intruders to find points of vulnerability in the system. This includes scanning firewalls, intrusion prevention systems, etc. to get a point of entry for the attack

- Weaponization

- Attackers develop malware by leveraging security vulnerabilities
- Attackers engineer malware based on their needs and the intention of the attack
- This process also involves attackers trying to reduce the chances of getting detected by the security solutions that the organization has in place

What Is the Cyber Kill Chain?

How does the Cyber Kill Chain Work? The cyber kill chain consists of 7 distinct steps:

- Delivery

- The attacker delivers the weaponized malware via a phishing email or some other medium.
- The most common delivery vectors for weaponized payloads include websites, removable disks, and emails.
- This is the most important stage where the attack can be stopped by the security teams

- Exploitation

- The malicious code is delivered into the organization's system.
- The perimeter is breached here. And the attackers get the opportunity to exploit the organization's systems by installing tools, running scripts, and modifying security certificates.
- Most often, an application or the operating system's vulnerabilities are targeted.
 - ✓ Examples of exploitation attacks can be scripting, dynamic data exchange, and local job scheduling

What Is the Cyber Kill Chain?

How does the Cyber Kill Chain Work? The cyber kill chain consists of 7 distinct steps:

- Installation

- A backdoor or remote access trojan is installed by the malware that provides access to the intruder
- This is also another important stage where the attack can be stopped using systems such as host-based intrusion prevention systems

- Command and Control

- The attacker gains control over the organization's systems and network
- Attackers gain access to privileged accounts and attempt brute force attacks, search for credentials, and change permissions to take over the control

- Actions on Objective

- The attacker finally extracts the data from the system
- The objective involves gathering, encrypting, and extracting confidential information from the organization's environment

How Can Cyber Kill Chain Protect Against Attacks?

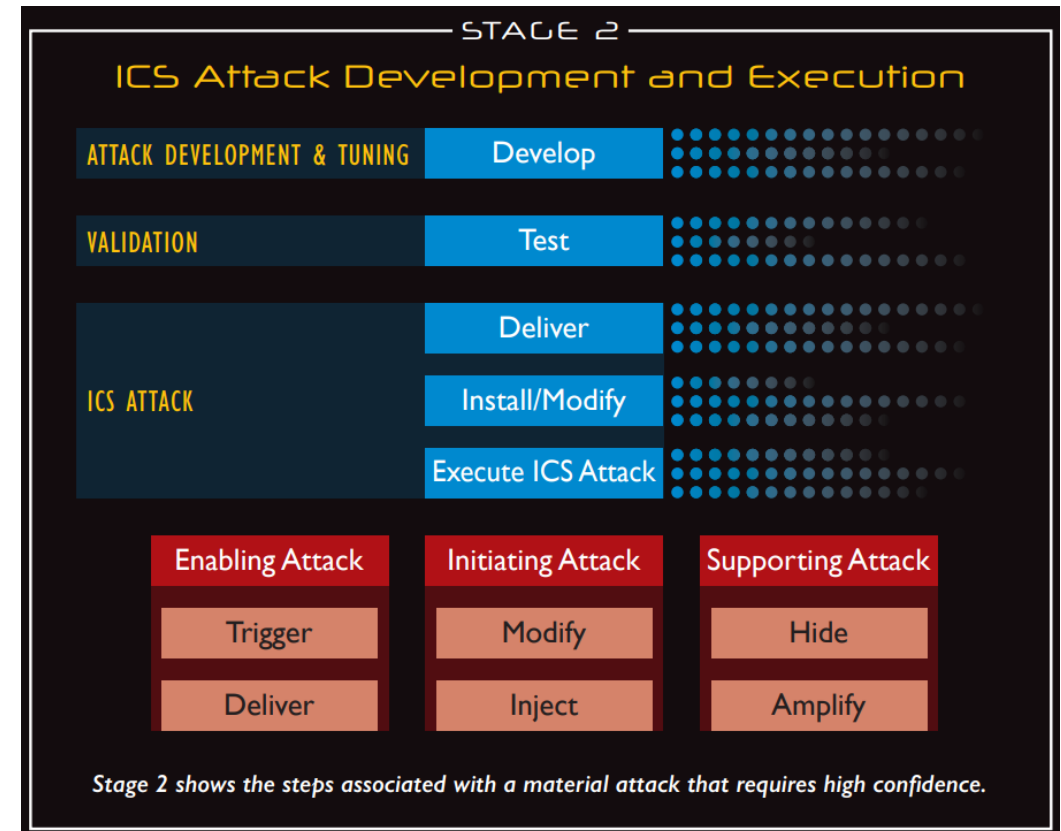
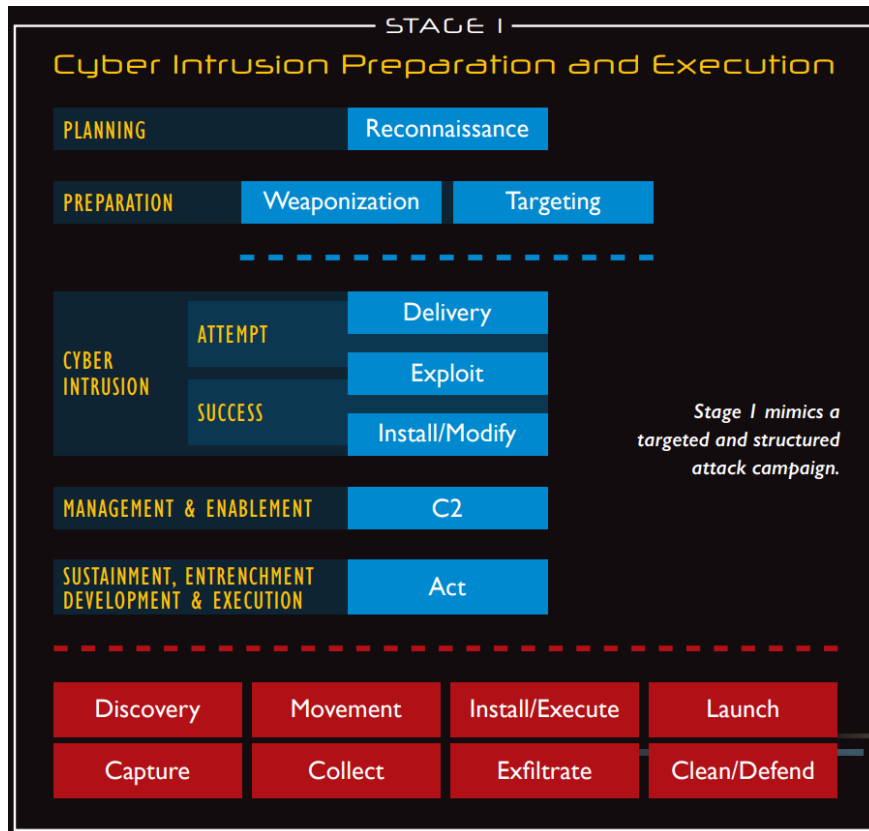
- Based on these stages, the following layers of control implementation are provided:
 - Detect
 - ✓ Determine the attempts to penetrate an organization
 - Deny
 - ✓ Stopping the attacks when they are happening
 - Disrupt
 - ✓ Intervene when the data communication is done by the attacker and stops it then
 - Degrade
 - ✓ This is to limit the effectiveness of a cybersecurity attack to minimize its ill effects
 - Deceive
 - ✓ Mislead the attacker by providing them with misinformation or misdirecting them
 - Contain
 - ✓ Contain and limit the scope of the attack so that it is restricted to only some part of the organization

How Can Cyber Kill Chain Protect Against Attacks?

- A cyber kill chain or cyber-attack simulation platform can be used by organizations to identify and fix the security gaps in their system within seconds
- Here's how simulating a cyber kill chain can protect against cybersecurity attacks:
 - **1. Simulate Cyber Security Attacks**
 - ✓ Real cyber security attacks can be simulated across all vectors to find vulnerabilities and threats. This includes simulating cyber attacks through email gateways, web gateways, web application firewall, and similar more
 - **2. Evaluate the Controls to Identify Security Gaps**
 - ✓ This involves evaluating simulations and identifying the areas of risk. Simulation platforms give you a detailed risk score and report around every vector
 - **3. Remediate and Fix the Cyber Security Gaps**
 - ✓ The next step is to fix the security gaps that were identified in the previous step.
 - ✓ This may include steps like installing patches and changing configurations to reduce the number of threats and vulnerabilities in the organization's system

Cyber Kill Chain for IT-OT Systems

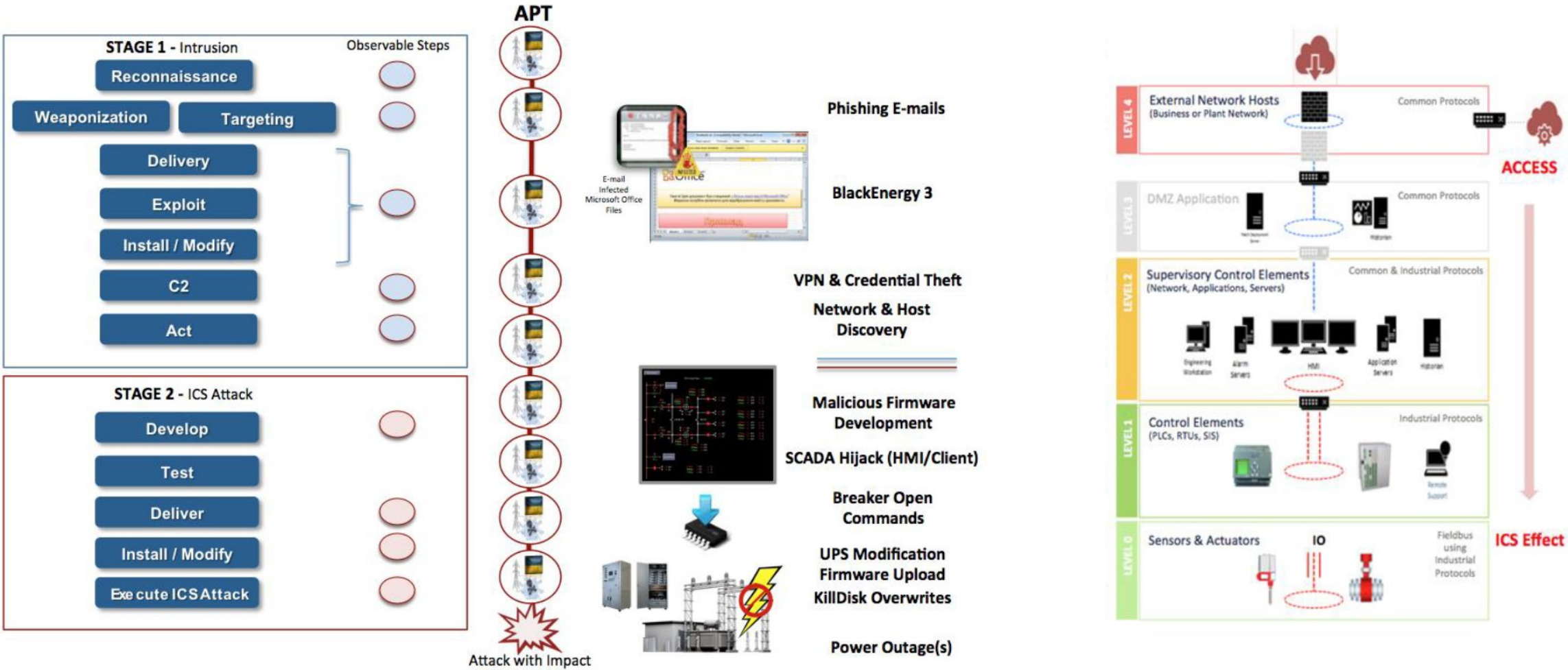
- Cyber Kill Chain Mapping proposed by ICS-SANS
- Steps an adversary takes to conduct an attack on Industrial Control System (ICS) such as SCADA



<https://www.sans.org/security-resources/posters/sliding-scale-cyber-security/105/download>

[Lee et al. Analysis of the Cyber Attack on the Ukrainian Power Grid, E-ISAC ICS SANS, 2016]

Cyber Kill Chain in Ukraine 2015 Attack



Defensive Measures for Cyber Security

TABLE I.4 An Overview of Typical Security Devices/Software

Name	Security check	Action taken
Firewall	TCP/IP packet header inspection	Block
Intrusion Detection System (IDS)	TCP/IP packet header and content inspection	Alert
Intrusion Prevention System (IPS)	TCP/IP packet header and content inspection	Block and alert
VPN: SSL/TLS	Authentication, encryption and integrity	Communication protection
VPN: IPsec	Authentication, encryption and integrity	Communication protection
Network access control (NAC)	Host health inspection, authentication, encryption and integrity	Access control

VPN: Virtual Private Network
SSL: Secure Sockets Layer
TLS: Transport Layer Security
IPsec: Internet Protocol Security

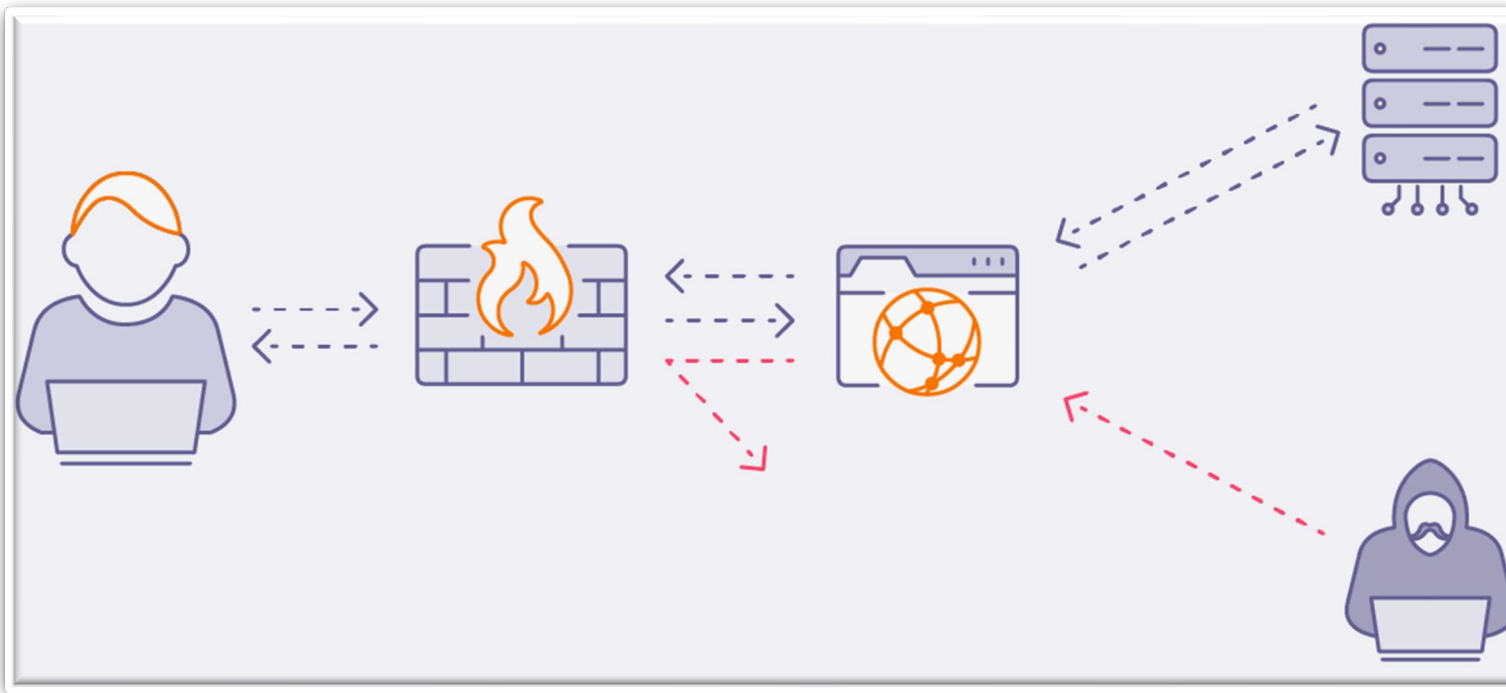
[Wu, Chwan-Hwa, & Irwin, J. David. (2013). Introduction to computer networks and cybersecurity. CRC Press.]

Firewall and Intrusion Detection / Prevention Systems (IDS/IPS)

- Perimeter defense by firewall, Intrusion Detection / Prevention System (IDS/IPS)
 - Place at the critical entry/exit points
 - Protect critical assets, such as a server farm, financial database
 - Use firewall, IDS/IPS
- Firewall
 - Computer host firewall protects a host
 - Network firewall protects the entrance to a network
 - Block packets based on IP address and port number in the header
 - Stateful inspection by maintaining a state transition table for a connection
- Host (computer / server) defense
 - Host IDS/IPS, Anti-Virus (AV) and host firewall
 - A.k.a. Endpoint defense

Firewalls

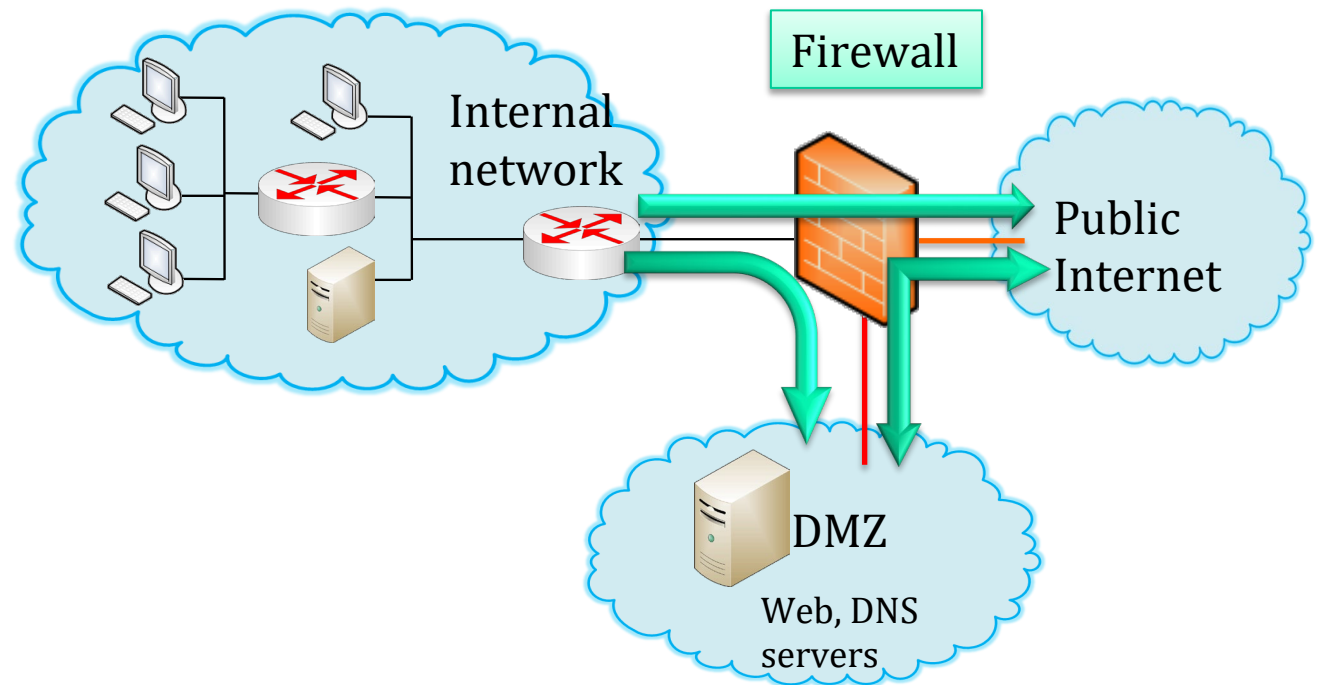
- Firewall is an application used to regulate access to an internal network from/to a public network (e.g., Internet)
- Firewall builds a blockade between an internal network that is assumed to be secure and trusted, and the public network, which is not



Source: www.avast.com

Firewalls

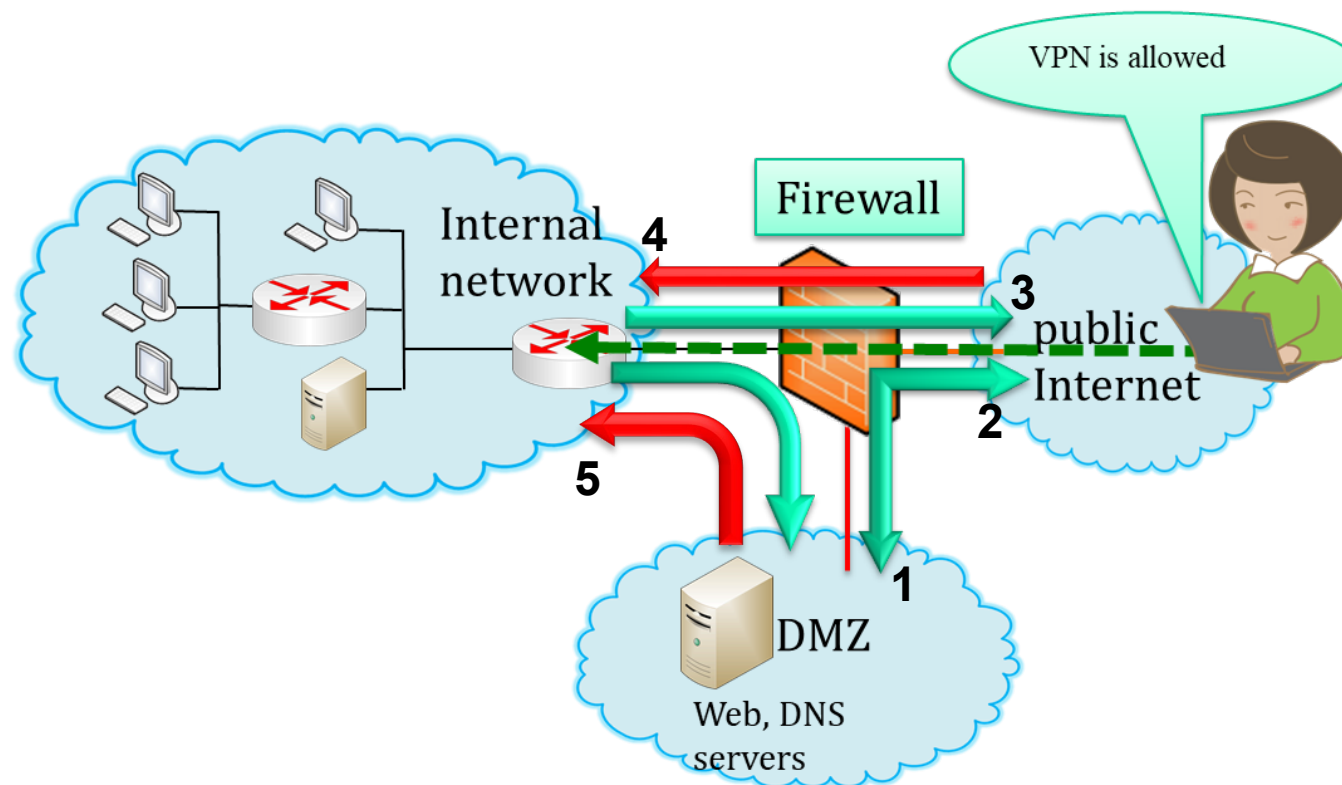
- Isolate organization's internal network (intranet) from Internet
- DMZ: demilitarized zone serving public information
- Allow:
 - Internet to DMZ
 - DMZ to Internet
 - Internal to Internet or DMZ
- Block:
 - Internet to Internal
 - DMZ to Internal except via VPN



Firewalls

- Firewall allows or blocks network traffic, based on IP addresses and port numbers

1. Allows connection from Internet to Demilitarized Zone (DMZ)
2. Allows connection from DMZ to Internet
3. Allows connection from Internal to Internet
4. Block connection from Internet to Internal
5. Block connection from DMZ to Internal



Firewalls Classification

- **Stateless packet filtering firewall:** compare each packet received to a set of established criteria, such as the allowed IP addresses, packet type, port number and other aspects of the packet protocol headers
- **Circuit-level gateway firewall:** monitor TCP handshakes and other network protocol session initiation messages across the network as they are established between the local and remote hosts to determine whether the session being initiated is legitimate or not
- **Application-level gateway firewall (proxy firewall):** filter packets not only according to the service for which they are intended (as specified by the destination port, IP address etc.) -- but also by other characteristics such as the HTTP request strings
- **Stateful inspection firewall:** state-aware devices not only examine each packet, but also keep track of whether it is part of an established TCP or other network session
- **Next generation firewall:** combines packet inspection with stateful inspection and includes some variety of Deep Packet Inspection (DPI)

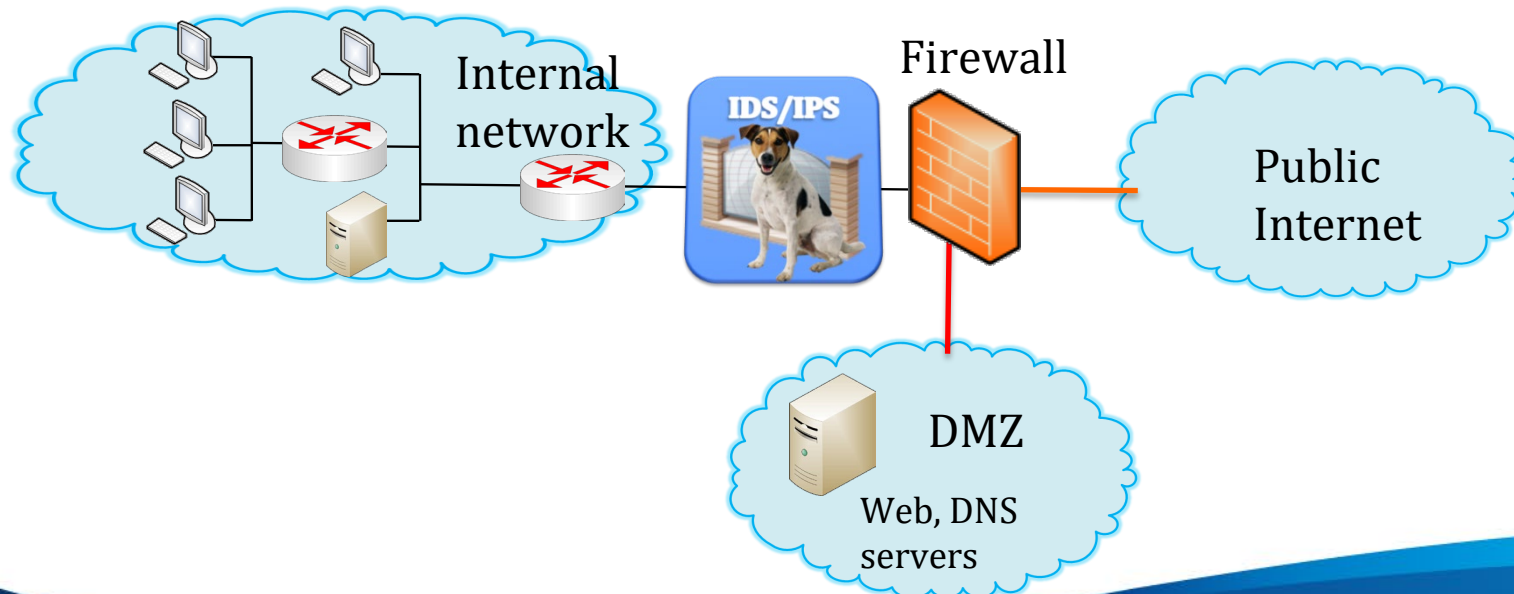
Firewalls Classification based on Generation

- First generation: included proxies and packet filters
- Second generation: featured Deep Packet Inspection (DPI)
- Third generation: Next Generation of Firewalls (NGFW) analyze layer 7 traffic
 - Capable of decrypting outbound Internet traffic and provide visibility into user browsing activity
- Fourth generation are cloud-based (virtual) firewalls

The Four Generations of Firewalls		
Generation	Capability	Attributes
First	Packet Filtering	Basic Network Policy
Second	Deep Packet Inspection	Application Identification
Third ("Next Gen")	Layer 7	User ID, Content Policy
Fourth	Cloud	Cloud Control Plane

Intrusion Detection / Prevention Systems

- IDS (Intrusion Detection System) / IPS (Intrusion Prevention System)
 - Host- or network-based IDS/IPS
 - Used for identifying or blocking suspicious packets at the entrances of critical assets
 - Basically, used to monitor potentially malicious traffic by inspecting the whole packet
 - IDS lets malicious packets pass, but sends alerts to a network administrator
 - IPS blocks malicious packets and sends alerts to a network administrator



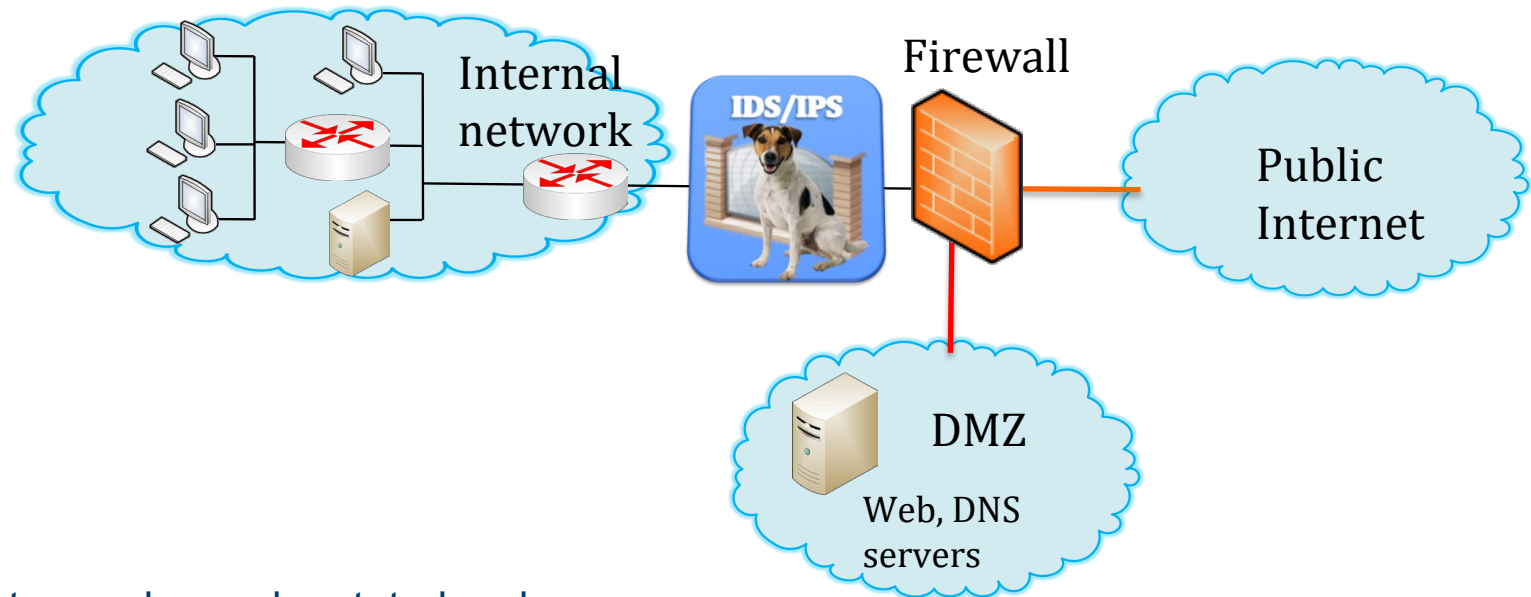
Intrusion Detection / Prevention Systems

- IDS/IPS can detect a wide range of attacks

- Network mapping (reconnaissance using nmap)
 - ✓ Port scans
 - ✓ TCP stack scans
- DoS bandwidth-flooding attacks
- Worms and Viruses
- Operating system vulnerability attacks
- Application vulnerability attacks

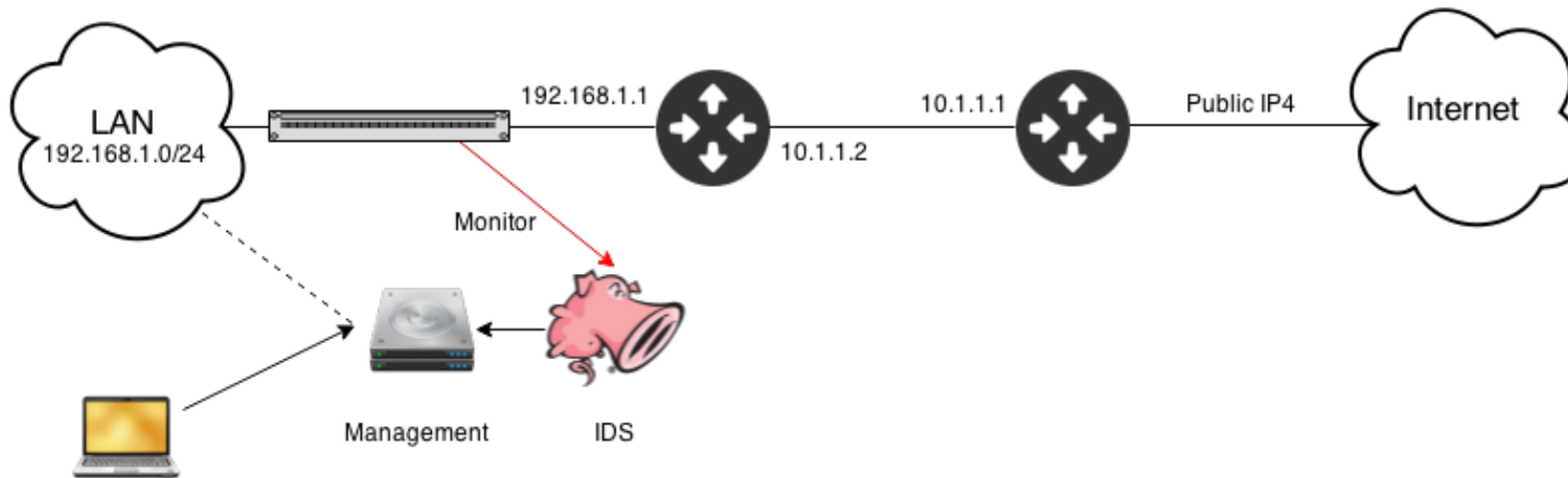
- Drawbacks:

- Signature-based detection cannot detect zero-day and mutated malware
- IDS: too many false positive alarms
- IPS: only blocks the packets that are sure to be malicious

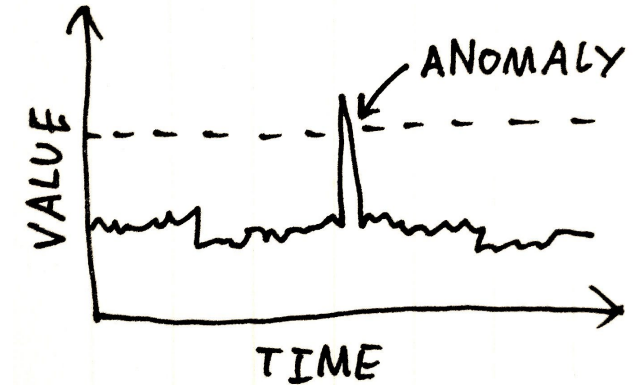


Snort

- Snort is a free and open-source network-based IDS/IPS
 - Most widely deployed intrusion detection and prevention technology worldwide
 - Works based on a large set of rules of known vulnerabilities
 - Combines benefits of signature and anomaly-based inspection methods
 - Community-maintained Snort rule sets constantly evolve for emerging threats



Source: techanarchy.net



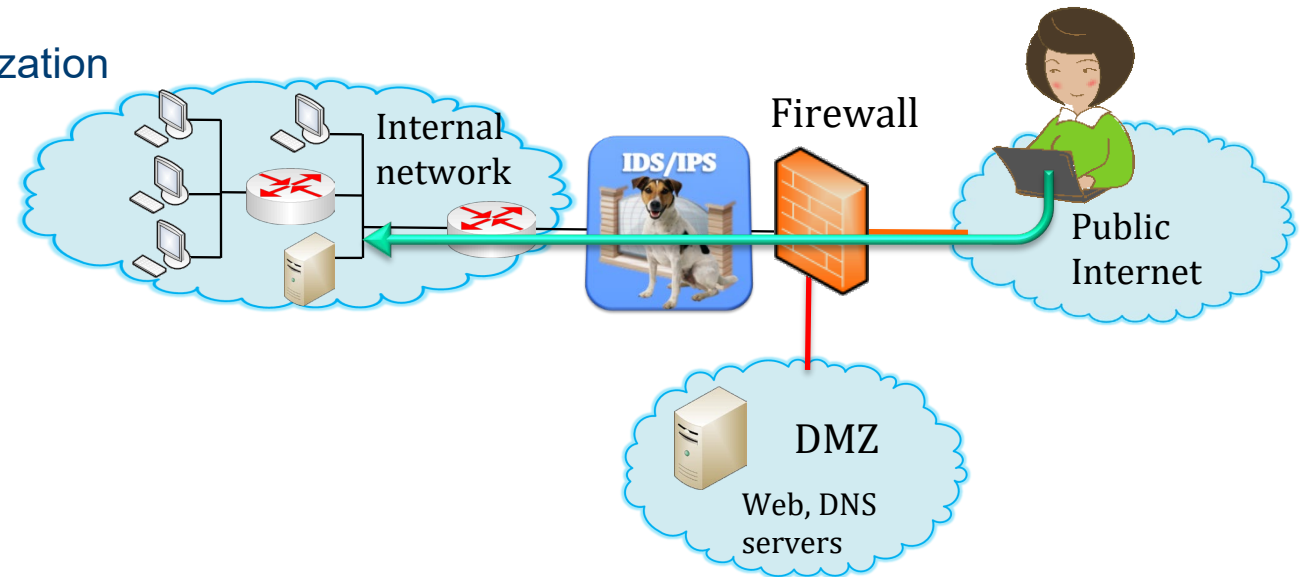
Source: towardsdatascience.com



Source: snort.org

Virtual Private Networks (VPN) and Access Control

- VPN provides a tunnel through the firewall
- Secure information transmission and access
 - Guarded by encryption, authentication and authorization
 - SSL/TLS: between session and transport layer
 - ✓ For Internet shopping and web mail
 - ✓ For VPN (virtual private network)
 - IPsec: network layer
 - ✓ VPN (virtual private network)



- Firewall allows SSL/TLS and VPN to pass through so that the external Internet can access internal resources

VPN: Virtual Private Network

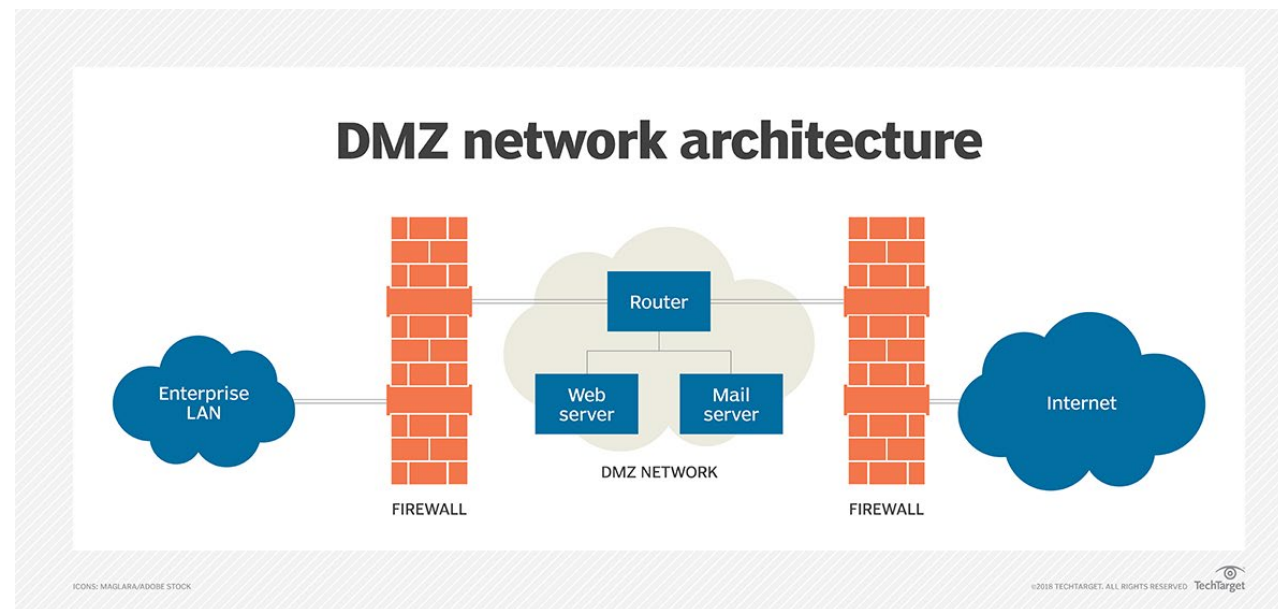
SSL: Secure Sockets Layer

TLS: Transport Layer Security

IPsec: Internet Protocol Security

Demilitarized Zone (DMZ)

- DMZ is a physical or logical subnet that separates LAN from other untrusted networks -- usually, the public Internet
 - DMZs are also known as perimeter networks
 - Any service provided to users on the public Internet should be placed in DMZ network
 - ✓ External-facing servers, resources and services are usually located there
 - ✓ Examples: web, email, domain name system, File Transfer Protocol and proxy servers





GET IN
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